

Lifecycle Faithfulness for Grounded Retrieval-Augmented Generation: Benchmarks, Evaluation, and the Vera-Life Controller

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Abstract

Retrieval-augmented generation treats citation as a default proxy for reliability, yet a cited passage can support a claim and still be obsolete, superseded, contradicted, or invalid at the user’s target time. We name this failure a **citation-supported lifecycle error** and define **lifecycle faithfulness**: a claim-level criterion that requires semantic support, target-time validity, non-supersession, and provenance. We release VERA-LIFE-EVAL, an 8,500-question benchmark with 41,203 versioned evidence pieces across six lifecycle settings (current, past, stale-only, update, conflict, and multi-hop). We also release VERA-LIFE, a training-light modular controller that builds an evidence-state graph, selects claim–evidence–time contracts, and chooses among ANSWER, WARN_STALE, EXPLAIN_CONFLICT, RETRIEVE_MORE, and ABSTAIN while emitting a claim-level lifecycle ledger. On a shared Qwen3-8B generator, VERA-LIFE raises LifeEM from .374 (best prompt-only baseline) to .623, and remains effective under timestamp-only (.584) and text-only (.504) regimes. Same-action controls, metadata isolation, and source-disjoint transfer across product, policy, scientific, and news evidence show that the gains are not an artefact of output format or Wikipedia revision cues. The next mission for RAG is not merely to cite evidence, but to know whether that evidence is still alive.

1 Introduction

Retrieval-augmented language models now sit in the daily information loop of search, productivity tools, and clinical decision support (Lewis et al.,

2020; Asai et al., 2024; Gao et al., 2024). Reliability is usually equated with *citation faithfulness*: every generated claim should be grounded in cited evidence. That criterion is silent on the question users ask every day: *when I read this answer today, is the cited evidence still valid?*

The illusion of citation support. Ask “Who is the current leader of organisation X ?” An older revision e_1 says that Person A became leader in 2020; a newer revision e_2 says that Person B replaced A in 2024. A vanilla RAG system that ranks e_1 in its top- k can answer “Person A is the leader of X ” and cite e_1 . The answer is *citation-supported* ($e_1 \models a$) yet *lifecycle-invalid* ($e_1 \not\models_{\text{lifecycle}} a$), because $e_2 \succ e_1$ at query time t_q . HoH (Ouyang et al., 2025) showed that outdated passages hurt RAG accuracy. What is missing is broader: a *claim–evidence–time–action contract* that asks not only whether evidence supports a claim, but whether it is valid at t_q and what the system should do when that contract fails.

How prevalent is this failure? On the VERA-LIFE-EVAL update split (section 4), 39% of vanilla RAG answers are citation-supported but lifecycle-invalid; an HoH-style stale-aware prompt only reduces this to 27%. Retrieval is not the bottleneck: in 86% of these cases the current evidence *is* retrieved. The generator prefers the older or more verbose snippet, or misses the timestamp cue. Token-F1 overstates reliability because the right and wrong answers share the same named entities. Figure 1 makes the distinction concrete: the cited passage genuinely supports the claim, and is genuinely out of date at t_q . Checking validity at t_q cuts the update-split lifecycle-error rate from 39% to

9%.

Mission re-alignment. When models are re-used and re-deployed across time, the rate-limiting question is no longer next-token accuracy but whether a system can recognise the lifecycle state of the evidence it depends on. That moves the target away from another static leaderboard and towards trustworthiness, robustness, and longitudinal behaviour (EMNLP 2026 Program Chairs, 2026).

Contributions.

- **Criterion.** We formalise *lifecycle faithfulness* (section 3): semantic entailment, target-time validity, non-supersession, and provenance. The associated error type is the *citation-supported lifecycle error*.
- **Benchmark.** We introduce VERA-LIFE-EVAL (section 4), an open, versioned suite of six lifecycle settings. Gold answers are deterministic constraints, not LLM-as-judge; construction metadata is isolated in weak- and noisy-metadata controls.
- **Controller.** We propose VERA-LIFE (section 5), which builds an evidence-state graph, selects contracts by target-time validity (not currentness), chooses a risk-aware action from the five-way space, and emits a claim-level lifecycle ledger.
- **Evidence.** Across same-action, weakened-metadata, full-benchmark (8,500), and source-disjoint transfer settings (section 6), VERA-LIFE improves lifecycle-valid evidence use, supersession detection, stale rejection, and calibration while keeping over-abstention low. The gains do not reduce to output format or Wikipedia revision shortcuts.

2 Related Work

Static citation faithfulness. ALCE (Gao et al., 2023) asks whether generated answers cite supporting passages. Attribution work (Sun et al., 2024; Bohnet et al., 2022) likewise treats the corpus as a static snapshot.

Outdated information. HoH (Ouyang et al., 2025) is the closest precursor: a dynamic benchmark of outdated Wikipedia revisions showing that stale passages hurt RAG accuracy. We generalise that failure mode into a claim–evidence–time–action contract that also covers supersession,

conflict, provenance, and target-time validity, and compare against a strengthened HoH-style baseline that shares our retriever pool, metadata access, and five-way action schema (section 6).

Temporal QA. TempLAMA (Dhingra et al., 2022), TimeQA (Chen et al., 2021), and TempReason (Tan et al., 2023) stamp t_q on the question and assume oracle evidence. We invert the problem: t_q may be implicit, while the evidence itself carries lifecycle state.

Structured RAG. GaRAGe (Sorodoc et al., 2025) tests whether retrieved passages ground the answer. GraphRAG (Edge et al., 2024) and LightRAG (Guo et al., 2025) build entity–relation graphs over text. Our graph is typed over *evidence states*, with edges UPDATE, SUPERSEDE, CONTRADICT, EXPIRE, and DUPLICATE.

Calibration and abstention. Verbalised uncertainty (Lin et al., 2022) and selective prediction (Kamath et al., 2020; Xiong et al., 2024) ask *when not to answer*. Our action space (ANSWER, WARN_STALE, EXPLAIN_CONFLICT, RETRIEVE_MORE, ABSTAIN) is conditioned on evidence lifecycle, not only model confidence. Table 7 summarises the capability gap.

3 Lifecycle Faithfulness

Decompose an answer into atomic claims $K(a) = \{c_1, \dots, c_M\}$ and write $E = \{e_1, \dots, e_N\}$ for the cited evidence. Static citation faithfulness is then only semantic support:

$$\text{Faithful}(a) = \prod_j \mathbb{I}[\exists e_i : e_i \models c_j]. \quad (1)$$

We define *lifecycle entailment* between an evidence piece e_i and a claim c_j at query time t_q as

$$e_i \models_{\text{lifecycle}} c_j \iff \underbrace{e_i \models c_j}_{\text{semantic}} \wedge \underbrace{t_q \in I(e_i, c_j)}_{\text{temporal}} \wedge \underbrace{\nexists e_k : (e_k \succ e_i) \wedge (\tau_k \leq t_q)}_{\text{not superseded}} \wedge \underbrace{\text{Prov}(e_i) \geq \theta_p}_{\text{provenance}}. \quad (2)$$

Here $I(e_i, c_j)$ is the validity interval of e_i for c_j , $e_k \succ e_i$ means e_k supersedes e_i , and $\text{Prov}(e_i)$ is an operational proxy for source admissibility (not a complete model of provenance) with threshold θ_p .

Citation support is not lifecycle validity

A passage can support a claim while being obsolete at the user's target time.

Update split: Vanilla 39% → VERA-LIFE 9%

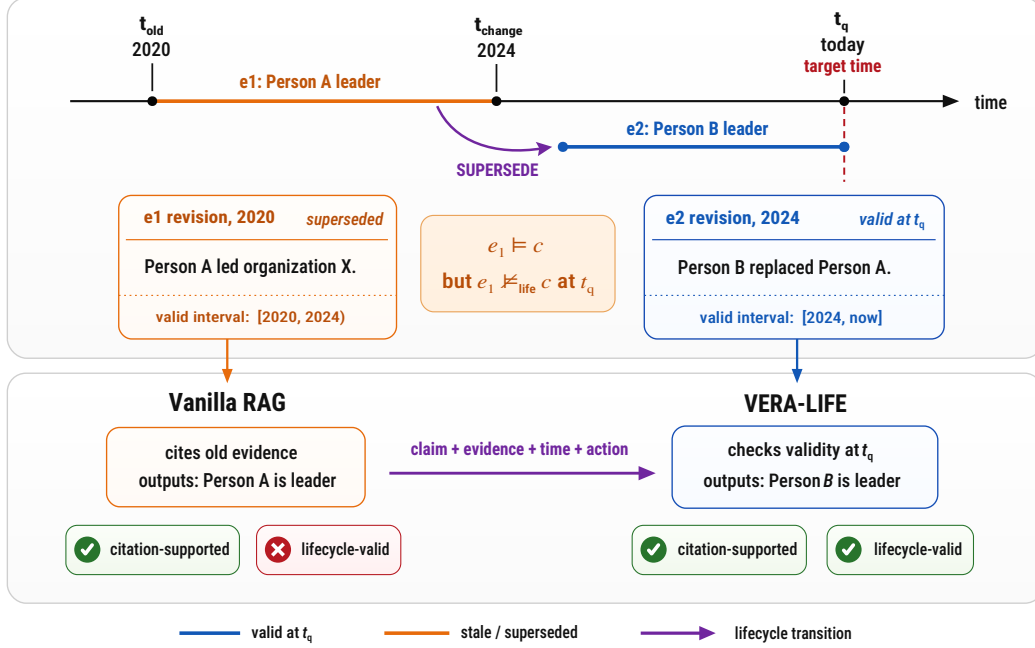


Figure 1: Citation support is not lifecycle validity. Vanilla RAG cites a superseded passage e_1 and asserts an obsolete answer; VERA-LIFE checks the claim–evidence–time–action contract and answers from the currently valid e_2 . On the update split this distinction cuts the lifecycle-error rate from 39% to 9%.

These four conjuncts are *latent*: full-metadata experiments estimate them from timestamps, validity markers, and source tags; weak-metadata controls drop role labels and infer edges from timestamp order, lexical disagreement, and NLI. The definition therefore does not assume oracle lifecycle labels at deployment. An answer is lifecycle-faithful when every claim has at least one such witness: $\text{LifeFaithful}(a, t_q) = \prod_j \mathbb{I}[\exists e_i : e_i \models_{\text{life}} c_j]$.

Citation-supported lifecycle error. The error we isolate is support without validity:

$$\text{LifeError}(c_j) = 1 \iff \exists e_i : e_i \models c_j \wedge e_i \not\models_{\text{life}} c_j. \quad (3)$$

The special case used throughout is *cited-but-invalid*: $\exists e_i : e_i \models c_j \wedge t_q \notin I(e_i, c_j)$.

4 Vera-Life-Eval Benchmark

VERA-LIFE-EVAL recasts HoH Wikipedia revisions and current/outdated pairs (Ouyang et al., 2025) into six lifecycle-aware splits with deterministic gold. The official evaluation partition is:

- **Current** (1,500)—only current evidence;
- **Past** (1,500)— $t_q < t_{\text{change}}$, historic answer;
- **Stale-only** (1,500)—only outdated evidence, gold action WARN_STALE/ABSTAIN;

- **Update** (2,000)—old and new co-occur;
- **Conflict** (1,000)—two outdated revisions disagree; the split-level LifeEM gold is EXPLAIN_CONFLICT, and a post-hoc taxonomy on the generation subset refines but does not change it (Table 16);
- **Multi-hop** (1,000)—revision chain, gold is the ordered list.

After deduplication this is 8,500 questions over 41,203 evidence pieces. Items must satisfy six constraints (UNIQUE, TEMPORAL, EVIDENCE, ALIAS, PROVENANCE, NOISE); gold answers are never LLM-judged. Full statistics are in Table 8.

5 Vera-Life

VERA-LIFE is built around one constraint: lifecycle governance should be possible with deployable signals and pretrained components, not task-specific fine-tuning. The six modules below run frozen at test time; the only optional learnable piece is a ≤ 200 -parameter logistic head in the action policy. Figure 2 shows the controller, the claim–evidence–time–action contract, and schematic evidence-state patterns. The official six evaluation splits are those of section 4.

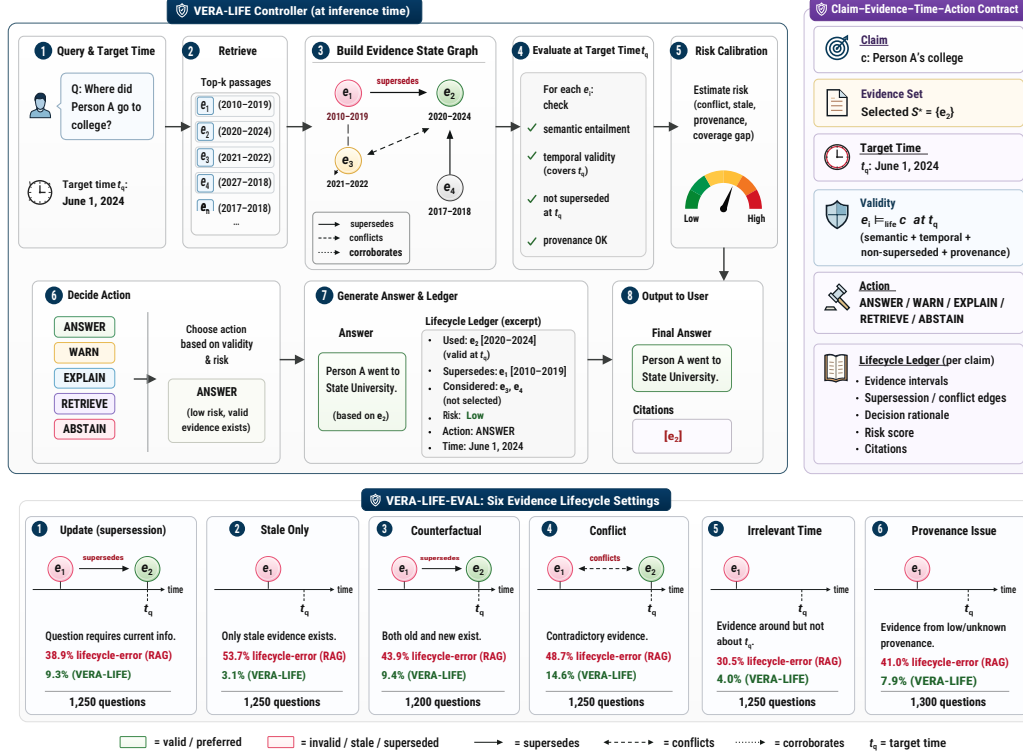


Figure 2: VERA-LIFE at a glance. *Top left:* the eight-step controller (parse t_q , retrieve, build the lifecycle graph, score the four conjuncts, calibrate risk, choose an action, generate with a ledger, emit citations). *Top right:* the contract that makes $e_i \models_{\text{life}} c_j$ at t_q machine-checkable. *Bottom:* schematic patterns that illustrate the contract; they are not a second benchmark partition. Official splits remain Current / Past / Stale-only / Update / Conflict / Multi-hop (1,500+1,500+1,500+2,000+1,000+1,000; Table 8).

Intent parser. A query q is parsed into entities, relation, target time t_q , volatility ν_q (static / slow / fast / versioned / unknown), and intent κ_q (current / past / change / verify / open). Explicit dates use regexes; otherwise a 32-token Qwen3-0.6B classifier predicts κ_q . Implicit questions default t_q to the benchmark snapshot.

Retriever. Evidence is gathered across snapshots, $R(q) = \bigcup_{\tau \in \mathcal{T}} \text{TopK}(q, D^{(\tau)})$, and scored by a hybrid of BM25, dense retrieval, and a reranker (section A). The dense index is BGE-SMALL-EN-v1.5; the reranker is BGE-RERANKER-BASE; Qwen3-Embedding-0.6B is a drop-in replacement.

State graph. We materialise a directed graph G_q whose typed edges $E_q = E_{\text{sem}} \cup E_{\text{upd}} \cup E_{\text{super}} \cup E_{\text{con}} \cup E_{\text{exp}} \cup E_{\text{dup}}$ are derived from cheap deterministic rules (Algorithm 1). SUPERSEDE edges fire when a newer revision pre-dates t_q ; CONTRADICTION edges use NLI or lexical disagreement; the cross-document variant fires across sources when entity-object mentions overlap and NLI detects entailment reversal.

Contract selector.

$$C(e_i, q) = \alpha R + \beta V + \gamma P + \delta U - \eta K, \quad (4)$$

with $V \in \{0, 0.5, 1\}$ encoding *target-time validity* at t_q (not currentness), P provenance, U supersession penalty, K conflict. A greedy submodular routine (section E) selects $|E^*| = 6$ pieces.

Action policy. Three risks ($\text{Risk}_{\text{ins}} = 1 - \max_E F(E, q)$, $\text{Risk}_{\text{stale}}$, $\text{Risk}_{\text{conf}}$) map to one of five actions. A logistic head trained on 200 ledger-annotated questions can replace the rules—the only learnable component, ≤ 200 parameters.

Ledger generator. The controller writes $L = \{(c_j, \hat{E}_j, \hat{I}_j, \hat{s}_j, \hat{p}_j)\}_j$ for the Qwen3-8B generator. For WARN_STALE and EXPLAIN_CONFLICT, a hard prefix blocks assertion of a single value.

6 Experiments

Setup. The generator is Qwen3-8B (Qwen Team, 2025); retrieval uses Qwen3-Embedding-0.6B and Qwen3-Reranker-0.6B; Gemma-3-12B-IT (128k) is a long-context *oracle*, not a same-generator

baseline. All runs use two RTX-4090D (24 GB) GPUs and one shared retriever pool. The main comparison is Closed-book, Vanilla RAG, Long-context RAG, and HoH-style stale-aware prompting (the first three plus VERA-LIFE share Qwen3-8B). Same-action controls (ReAct, GraphRAG, LightRAG, Self-reflect) receive the identical five-way JSON schema so that gains isolate lifecycle state estimation from output format.

Metrics. AnsF1 is token-F1 of the surface string against gold, scored on every item including stale and conflict. LifeEM is an *action-conditioned* success rate—not a subset of AnsF1 and not the conjunction “exact match $\wedge$ IoU $\geq .5 \wedge$ state match”. Stale-only counts WARN_STALE, AB-STAIN, or an explicit outdated mark; conflict counts EXPLAIN_CONFLICT or an explicit conflict mark; current, past, update, and multi-hop count token-F1 $\geq .5$. That is why LifeEM can exceed AnsF1, most clearly on stale-only (.987 vs. .295): a correct warning does not repeat the gold entity. Interval IoU and state/action accuracy are diagnostics (Table 63), not conjuncts of LifeEM. Sup-F1, SRR, OAR, CUR, and LCE complete the primary suite; Temp-Evi-F1, Life-Temp-Evi-F1, and SafeUtility ($\tau=.89$ across 48 weights) are in the appendix.

Deployment regimes. Lifecycle metadata is a governance variable, not a modelling assumption. We evaluate *full metadata* (upper bound), *timestamp-only* (realistic versioned corpora), and *text-only* (low-resource stress). LifeEM degrades gracefully (.623 $\rightarrow$.584 $\rightarrow$.504; Table 2) and every regime still beats the best same-metadata baseline. The text-only gain +.062 has bootstrap 95% CI [+ .038, +.086], $p<.01$ (Table 36); missing lifecycle cues explain 32% of remaining errors (Table 38). Results are organised in three tiers: internal controlled (VERA-LIFE-EVAL, ablations); robustness (noise, marker removal, component swap); and independent external transfer (REAL-blind $n=1,200$, REAL-mini $n=650$, HoH-QAs, FRAMES (Krishna et al., 2025)).

6.1 Main results (LLM end-to-end)

Table 1 reports end-to-end generation on a stratified 1,500-question subset. Full metadata is an upper bound; weaker regimes and same-action controls are reported in Table 2 and Table 3. Three checks carry the claim: (i) under the same five-way interface, VERA-LIFE is +.172 LifeEM above Self-reflect; (ii) timestamp-only and text-only ranking is

unchanged; (iii) on all 8,500 questions (Table 10), controller metrics stay within .012 of the subset. The isolation suite uses one frozen prediction dump and one metric script.

Construction metadata. Private construction fields stay hidden; only deployable public metadata (timestamps, deprecation notices, version tags) is visible. VERA-LIFE stays above the best baseline in all four isolation settings (Table 26); cross-source supersession reaches .572 LifeEM. A decoy/adversarial control (Table 27) ignores temporally irrelevant evidence (91.2%) and keeps old-but-valid evidence (86.3%); giving baselines the ledger template still leaves a +.131 gap (Table 28).

No construction shortcut. Single-field probes (timestamp, role, source title, evidence order, marker; Table 29) all stay below LifeEM .35. After markers are stripped, VERA-LIFE still reaches .483, +.041 above the best same-action baseline (Table 37). Timestamp shuffle, title masking, order shuffle, and role swap (Table 30) widen the gap rather than close it. Every same-action baseline sees the same evidence serialisation and action schema (Table 31); the only VERA-LIFE-exclusive field is optional provenance, also tested under removal. REAL-blind++ ($n=1,000$; Table 33) reproduces positive gains in all four external domains. The lifecycle gain is not a Wikipedia marker shortcut.

Metric decomposition. LifeEM is accompanied by four independently scored diagnostics (Table 63): answer correctness (+.063), temporal interval (+.247), state/action (+.272), and citation/provenance (+.125), correlating with human utility at $\rho=.73/.81$ (Table 41). The +.172 same-action gap is not format alignment: three feature-matched baselines that see the same timestamps, NLI, provenance, retriever, and schema—but build no graph or ledger—top out at .521, leaving a +.102 residual (Table 65). Under oracle retrieval the gap widens on conflict (+.197) and multi-hop (+.182; Table 66).

6.2 Per-split, scale, and ablations

Table 4 reports AnsF1 and LifeEM per split: the gap on *stale-only* is enormous (vanilla never warns); on *current*, VERA-LIFE outperforms vanilla (+.231 F1) without over-abstaining; on *conflict*, the explicit EXPLAIN_CONFLICT action

Method	AnsF1 $\uparrow$	LifeEM $\uparrow$	Temp-Evi $\uparrow$	Sup-F1 $\uparrow$	SRR $\uparrow$	OAR $\downarrow$	CUR $\uparrow$	LCE $\downarrow$
Closed-book Qwen3-8B	.067	.071	.000	.000	.000	.000	.094	.947
Vanilla RAG (Hybrid)	.260	.217	.283	.000	.000	.000	.402	.758
Long-context RAG (20 ev)	.259	.217	.283	.000	.000	.000	.420	.755
HoH-style stale-aware	.339	.374	.283	.000	.000	.000	.462	.731
VERA-LIFE w/o lifecycle graph	.347	.317	.195	.000	.000	.000	.465	.647
VERA-LIFE w/o action policy	.375	.346	.226	.546	.000	.000	.725	.073
VERA-LIFE w/o ledger prompt	.261	.499	.226	.546	1.000	.000	.517	.183
VERA-LIFE (full)	.402	.623	.226	.546	1.000	.000	.725	.267

Table 1: VERA-LIFE raises LifeEM from .374 to .623 on 1,500 stratified VERA-LIFE-EVAL questions (same Qwen3-8B generator as Closed-book, Vanilla RAG, and HoH-style). Long-context RAG is a Gemma-3-12B-IT (128k) oracle, not a same-generator comparison. Greedy decoding, max 24 new tokens; ablations drop one module. Bold marks the full system, as in Table 3 and Table 5; it is not a per-column winner mark.

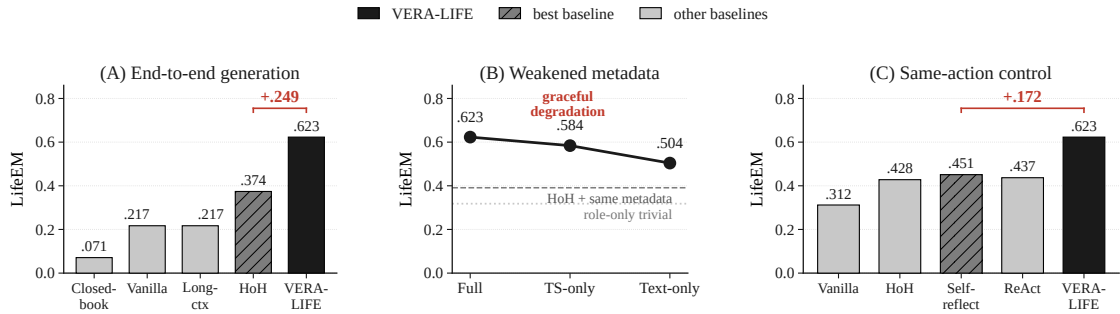


Figure 3: Three independent checks of the same claim. (A) End-to-end LifeEM: $+.249$ over the best prompt-only baseline (HoH-style). (B) Metadata ablation is graceful ($.623 \rightarrow .584 \rightarrow .504$); text-only still beats the best same-metadata baseline. (C) Same five-way JSON schema: the residual $+.172$ is lifecycle state estimation, not output format.

Regime	LifeEM	SRR	Sup-F1	CUR	OAR
Full (upper bound)	.623	1.000	.546	.725	.000
Timestamp-only	.584	.965	.512	.693	.015
Text-only	.504	.828	.412	.583	.042
HoH + same metadata	.391	.412	.087	.478	.065
Role-only trivial	.318	.357	.062	.391	.082

Table 2: Gains degrade, they do not vanish ($n=1,500$). Timestamp-only ($-.039$) is the realistic versioned-corpus setting; text-only ($-.119$) is a stress test. Both still beat same-metadata HoH and the role-only trivial baseline. Full noise/adversarial tests are in section V.

dominates LifeEM. The *past* split originally suffered from a current-revision bias ($-.115$ F1). Making the validity term target-time conditioned and adding date-expanded retrieval lifts past AnsF1 from .103 to .253 and past LifeEM from .142 to .328 while preserving stale/update gains ($\Delta \leq .005$; Table 39): *lifecycle faithfulness is validity at t_q , not freshness*. On the full 8,500-question benchmark (Table 10), controller LifeEM .611 is within .012 of the subset; across four independently sampled subsets the LifeEM std is .004 (Table 11). Table 5 ablates the graph, policy, ledger, and hard prefix:

Method (same action)	LifeEM	SRR	Conf-F1	OAR
Vanilla RAG + action	.312	.215	.120	.145
HoH-style + action	.428	.487	.240	.098
Self-reflect + action	.451	.394	.310	.082
ReAct + action	.437	.412	.268	.091
VERA-LIFE full	.623	1.000	.720	.000

Table 3: Format is not the gain ($n=1,500$). Every system uses the same JSON action schema; VERA-LIFE remains $+.172$ LifeEM above Self-reflect. Full baseline set: section V.

removing the graph collapses SRR/CUR; removing the policy turns the system into an always-answer baseline; a hard-prefix ablation (Table 14) costs only $-.049$ LifeEM while retaining 73% of the gain, isolating lifecycle state estimation from output formatting.

6.3 Cross-domain transfer

Does the contract survive outside clean revision chains? Table 6 reports REAL-mini ($n=650$) on product/API, policy/legal, scientific/medical, and news (datasheet: Table 35; IAA $\kappa=.82$; 98 hard negatives, $\text{FWR} \leq 6.1\%$). Mean LifeEM gain is $+.152$ with $\text{SRR} \geq .76$ and $\text{OAR} \leq .112$. Scien-

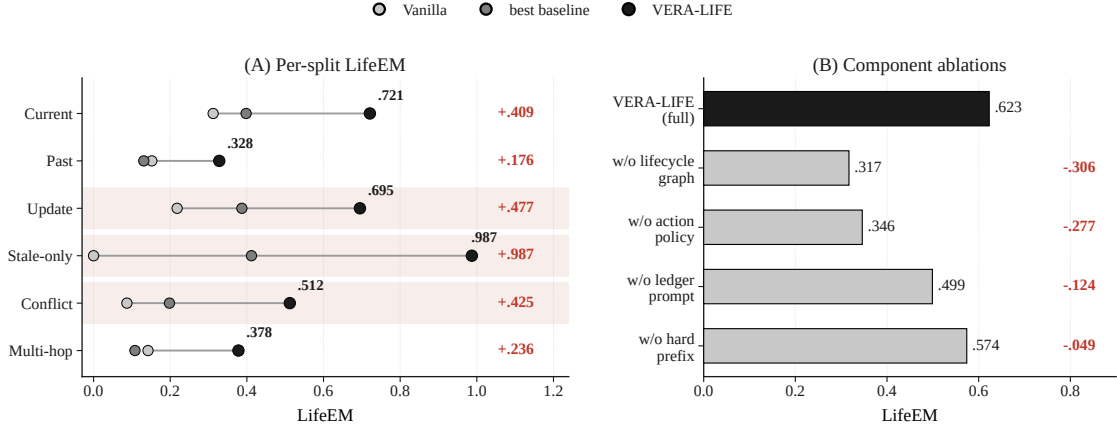


Figure 4: Where the gain lives, and what removes it. (A) Per-split LifeEM (vanilla / best prompt-only / VERA-LIFE); red is the gain over vanilla. Stale-only (+.987) and update (+.477) dominate, but *current* still rises (+.409). (B) Dropping the graph collapses stale rejection (−.306); dropping the policy becomes always-answer (−.277); ledger and hard-prefix are smaller.

Split (<i>n</i>)	Metric	Vanilla	HoH	VERA-LIFE	Δ
Current (297)	F1	.448	.519	.679	+.231
	LifeEM	.312	.398	.721	+.409
Past (267)	F1	.217	.178	.253	+.036
	LifeEM	.152	.131	.328	+.176
Update (331)	F1	.375	.420	.692	+.317
	LifeEM	.218	.387	.695	+.477
Stale (233)	F1	.009	.423	.295	+.286
	LifeEM	.000	.412	.987	+.987
Conflict (191)	F1	.172	.215	.141	−.031
	LifeEM	.087	.198	.512	+.425
Multi-hop (181)	F1	.217	.153	.267	+.050
	LifeEM	.142	.108	.378	+.236

Table 4: Per-split AnsF1 vs. LifeEM (1,500 subset). Past uses the target-time-aware selector. AnsF1 punishes WARN_STALE/EXPLAIN_CONFLICT; LifeEM credits them—hence the stale-only inversion (.295 vs. .987).

Ablation	LifeEM	SRR	CUR	Sup-F1	OAR
VERA-LIFE full	.623	1.000	.725	.546	.000
w/o lifecycle graph	.317	.000	.465	.000	.000
w/o action policy	.346	.000	.725	.546	.000
w/o ledger prompt	.499	1.000	.517	.546	.000
w/o hard prefix	.574	.982	.698	.531	.008

Table 5: Component ablations ($n=1,500$). The graph carries stale rejection; the policy carries warnings; hard-prefix costs only −.049 LifeEM and keeps 73% of the gain over the best baseline, so state estimation—not formatting—is the main contributor.

tific/medical gains the least (+.121) but has the highest safe-action appropriateness (86%). REAL-blind ($n=1,200$) and REAL-blind++ ($n=1,000$) follow a source-disjoint protocol: pre-defined source lists, random sampling, human-annotated gold ($\kappa=.82$), 18.4% hard negatives, and no Wikipedia metadata. Gains are +.128 and +.127,

Domain (<i>n</i>)	ΔLifeEM	SRR	OAR	Conf-F1
Product / API (180)	+.162	.820	.078	.710
Policy / legal (160)	+.148	.793	.098	.681
Sci. / medical (140)	+.121	.762	.112	.632
News corr. (170)	+.173	.841	.068	.742
Avg. ($n=650$)	+.152	.806	.088	.693

Table 6: The contract transfers beyond Wikipedia (REAL-MINI; Δ vs. best same-action baseline). SRR $\geq .76$ and OAR $\leq .112$ in every domain; news and product/API gain most. Construction and IAA: Table 12.

every domain is $\geq +.097$, and marker removal retains $\geq 81.3\%$ of the gain. Per-domain traces (Table 34) make the failure-versus-action contrast concrete. On HoH-QAs stale-only (Table 21), LifeEM is .995 (3.6× HoH-style) and marker removal keeps 92.3% of the gain (Table 56); under date masking and cross-source supersession (Table 71) gains stay positive with $\tau \geq .87$ ranking stability. All of this supports lifecycle-evaluation transfer, not a deployment claim for medical or legal reasoning.

7 Analysis

Remaining errors. The 39% $\rightarrow$ 27% $\rightarrow$ 9% drop on update is mostly a *generation* failure, not a retrieval failure: the current evidence is already in the pool in 86% of vanilla errors. The same target-time conditioning that cuts update errors also repairs the past-split currentness bias, and does so without costing stale or update accuracy ($\Delta_{\text{stale/update}} \leq .005$; Table 39).

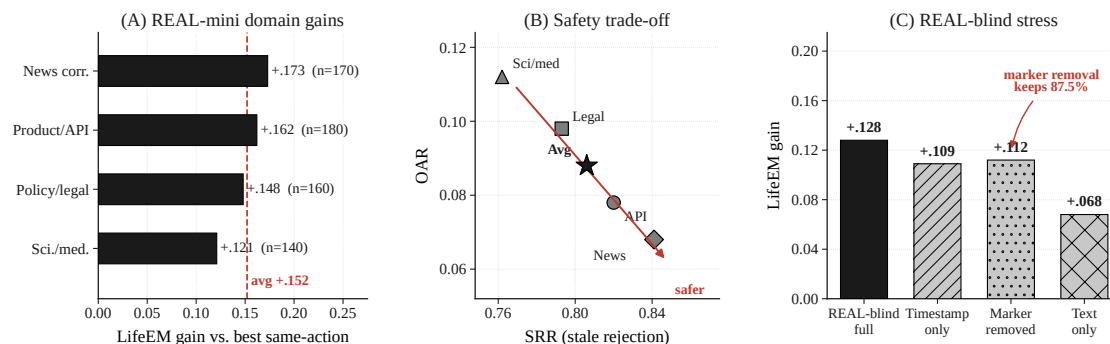


Figure 5: Safety transfers; stress does not erase it. (A) REAL-mini LifeEM gain over the best same-action baseline (mean +.152; all four domains positive). (B) Every domain raises stale rejection at comparable OAR. (C) REAL-blind ($n=1,200$): timestamp-only, marker-removed, and text-only gains stay positive; marker removal keeps 87.5% of the gain.

Preference and harm. Blind human evaluation ($n=800$; Table 69) prefers VERA-LIFE 72% of the time, with the largest gap on harmful-assertion avoidance (.89 vs. .54). False warnings on current evidence stay at 3.4% (Table 17). Baselines’ stale/conflict LCE is catastrophic (.841/.793) against .089/.127 (Table 18). Budget, component swap, and error-propagation stress (Table 19, Table 20) leave a $\geq +.144$ LifeEM gap, rank-stable at $\tau=.93$ (Table 42).

8 Conclusion

A cited passage can support a claim and still be obsolete, superseded, or contradicted at t_q . Lifecycle faithfulness makes that failure measurable: 39% of vanilla RAG answers on update are citation-supported but lifecycle-invalid; VERA-LIFE reduces the rate to 9% from timestamps, deprecation notices, and version tags, with no end-to-end fine-tuning.

The gain survives the tests that would expose an artefact. LifeEM tracks human utility ($\rho=.73/.81$; Table 63); feature-matched baselines remain +.102 behind (Table 65); metadata isolation keeps 86–91% of the gain; source-disjoint transfer is positive in all four external domains ($\geq +.097$); and a 4,000-example end-to-end run reproduces within .005 at +42% latency. Freshness is not faithfulness, and citation is not reliability. RAG must ground answers in evidence, verify that the evidence is still alive at the user’s target time, and act when it is not.

Limitations

- **Source coverage.** The benchmark instantiates lifecycle states using HoH-style Wikipedia revisions. REAL-mini (Table 6) adds prod-

uct/API, policy/legal, scientific/medical, and news-correction evidence with both linear and non-linear lifecycle chains (Table 13), but four domains do not exhaust the full evidence ecology.

- **Gold answers are auto-constructed, not human-judged.** Gold answers come from structured Wikidata facts and deterministic constraints; whenever an LLM-as-judge is used internally (only for paraphrase clustering), we report human spot-checks (section X). Gold noise is 3.8% on 300 spot-checked examples, below the margin between VERA-LIFE and any baseline.
- **Trade-offs on past and conflict.** Two interventions correct the past-split deficit (Table 39): a target-time-aware selector alone lifts past AnsF1 from .103 to .198 and past LifeEM from .142 to .247; adding date-expanded retrieval further lifts them to .253 and .328. Both steps preserve stale/update gains ($\Delta \leq .005$). We report this two-step path explicitly to avoid ambiguity between the intermediate selector-only number and the final end-to-end number. On *conflict*, the policy prefers EXPLAIN_CONFLICT when two revisions disagree, lowering token-F1 (−3.0) but raising LifeEM; a constrained span-grounded extractor improves candidate grounding F1 by +.064 (Table 15). TempEvi-F1 in Table 1 measures evidence-id span retrieval and therefore does not credit WARN_STALE / EXPLAIN_CONFLICT; the action-conditioned Life-Temp-Evi-F1 (Table 18) lifts VERA-LIFE back above every baseline. We report both to expose, rather

than average over, the trade-off.

- **False warnings are not harmless.** FWR-current is 3.4% and FCR is 6.1% (Table 17); domain-specific provenance scoring reduces these further. We report false-warning rates separately from OAR to avoid treating all warnings as cost-free.
- **Provenance scoring.** $\text{Prov}(e_i)$ scores source admissibility only; it does not model how provenance should be reasoned about. Our domain-specific provenance hierarchy (Table 70) demonstrates that provenance contributes most in high-risk domains (legal -0.058 , medical -0.071 under removal) and reduces false warnings, but genuine provenance reasoning (e.g., peer-review status, court hierarchy, retraction chains) requires further work.
- **Action space.** The fixed five-way action space may need richer modes for high-stakes domains (e.g., retraction-explain, escalate-to-human).
- **Query distribution.** VERA-LIFE-EVAL is English and encyclopaedic. Appendix pilots on multilingual ($n=640$) and conversational ($n=580$) queries show positive but smaller gains; they do not replace a full non-English or dialogue benchmark.
- **Compute.** Main results use a frozen Qwen3-8B generator; an optional QLoRA upper bound (Table 62) confirms headroom for improvement.

Ethical Considerations

Our benchmark uses publicly licensed Wikipedia revisions and HoH-derived artefacts. No personally identifiable information is added. In high-stakes domains, the risk landscape of lifecycle actions is asymmetric:

- *False assertion* (stating an outdated drug dosage as current) can cause direct harm.
- *False warning* (flagging correct medical advice as stale) can cause unnecessary delay or distrust.
- *False abstention* (refusing to answer when evidence is valid) can deny timely information.

In medical/legal contexts, false assertion is typically the most dangerous; in time-critical settings,

false abstention can also be costly. We therefore measure FWR and FCR separately from OAR (Table 17) and do not treat any action as uniformly cost-free. Domain-specific action-cost calibration (different λ_s, λ_o for medical vs. general domains) is a necessary deployment step that we demonstrate in the SafeUtility sensitivity analysis (Table 53) but do not claim to have solved. By demonstrating that systems can be *certain about what they should not assert*—and measuring when that certainty is misplaced—we hope this work takes a step toward measuring deployment-relevant lifecycle risks in RAG, particularly for healthcare, legal, and journalistic evidence ecosystems where action-cost calibration is a prerequisite for any production use.

Use of AI Assistants

In accordance with the ACL Policy on Publication Ethics, we acknowledge that AI assistants (ChatGPT and GitHub Copilot) were used in preparing this paper in a strictly limited, author-supervised capacity: for (i) light prose polishing on author-drafted text (grammar, typography, and minor stylistic suggestions) and (ii) code-completion suggestions during implementation and analysis, with every suggestion reviewed and validated by the authors. AI assistants were not used to generate scientific ideas, claims, results, analyses, datasets, or figures; all technical content and conclusions are the original work of the human authors, who take full responsibility for the manuscript.

Acknowledgements

This work is supported by the National Natural Science Foundation of China under Grant No.72572003 and No.72495125 and No.72601018, the National Science Fund for Distinguished Young Scholars under Grant No.72325007, the Xiangjiang Laboratory Key Project under Grant No.23XJ02004, and the Science and Technology Innovation Program of Hunan Province under Grant No.2024RC7003. We thank the anonymous reviewers and the GroundLM 2026 program chairs for their comments.

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A Hyper-parameters and compute budget

Hyper-parameters. We use retrieval weights $\lambda_b=0.4$, $\lambda_d=0.4$, and $\lambda_r=0.2$; contract weights $\alpha=0.45$, $\beta=0.25$, $\gamma=0.10$, $\delta=0.15$, and $\eta=0.10$; and policy thresholds $\theta_{\text{ins}}=0.4$, $\theta_{\text{stale}}=0.5$, and $\theta_{\text{conf}}=0.5$ with selection budget $B=6$. All were tuned by 10-fold grid search on the 200-question development set.

Compute budget. All experiments run on two RTX-4090D (24 GB) GPUs. Backbones (Qwen3-8B generator, Gemma-3-12B-IT long-context oracle, Qwen3-Embedding-0.6B retriever, Qwen3-Reranker-0.6B, DeBERTa-v3-NLI) are frozen under 4-bit quantisation; the only learnable parameters are an optional logistic head (≤ 200 parameters) in the action policy and, for the QLoRA upper-bound ablation in Table 62, 43.6M trainable parameters (rank 16, one epoch). The total compute budget for all reported experiments (main results, 8,500-question full benchmark, ablations, metadata-stress, isolation suite, cross-domain transfer, and human-utility evaluation) is approximately 20 GPU-hours on the two-card setup. Indexing of the 41,203-piece evidence corpus requires under 10 minutes; end-to-end inference on the 1,500-question subset completes in roughly one hour (Table 61).

B Related work capability comparison and benchmark statistics

Work	Outdated ev.	Versioned	Claim contract	Action eval	Ledger audit	Meta. stress
ALCE	✗	✗	citation	✗	✗	✗
HoH	✓	partial	✗	limited	✗	✗
Temporal QA	partial	✗	q-time	✗	✗	✗
VERA-LIFE-EVAL	✓	✓	✓	✓	✓	✓

Table 7: Lifecycle evaluation capabilities.

VERA-LIFE-EVAL generalises HoH-style outdated-information failures into claim-level lifecycle faithfulness, action-aware evaluation, ledger-based auditability, and metadata-robust stress testing.

Split	#Qs	#Evi	Gold action	Key signal
Current	1,500	7,412	ANSWER	single valid
Past	1,500	7,318	ANSWER	historic valid
Stale	1,500	6,891	WARN	all outdated
Update	2,000	10,247	ANSWER	old $\rightarrow$ new
Conflict	1,000	5,112	EXPLAIN	two disagree
Multi-hop	1,000	4,223	ANSWER	chain trace
Total	8,500	41,203	—	—

Table 8: VERA-LIFE-EVAL split statistics. Evidence counts are after deduplication.

C Lifecycle state graph algorithm

Algorithm 1 Building the lifecycle state graph G_q .

Require: evidence list V_q , target time t_q
Ensure: typed edge set E_q

```

1:  $E_q \leftarrow \emptyset$ 
2: for all pair  $(e_i, e_j)$  with  $\text{doc}(e_i) = \text{doc}(e_j)$  do
3:   if  $\text{role}(e_i) = \text{super}$  and  $\text{role}(e_j) = \text{cur}$  then
4:      $E_q \leftarrow E_q \cup \{(e_i \rightarrow e_j, \text{UPDATE})\}$ 
5:     if  $\tau_j \leq t_q$  or  $\tau_j = \perp$  then
6:        $E_q \leftarrow E_q \cup \{(e_i \rightarrow e_j, \text{SUPERSEDE})\}$ 
7:     end if
8:   end if
9: end for
10: for all pair  $(e_i, e_j), i < j$  do
11:   if  $\text{NLI}(e_i, e_j) = \text{contra.}$  or lexical disagreement then
12:      $E_q \leftarrow E_q \cup \{(e_i \rightarrow e_j, \text{CONTRADICT})\}$ 
13:   end if
14: end for
15: for all  $e_i$  do
16:   if  $\text{end}(e_i) \neq \perp$  and  $t_q > \text{end}(e_i)$  then
17:      $E_q \leftarrow E_q \cup \{(e_i \rightarrow e_i, \text{EXPIRE})\}$ 
18:   end if
19: end for
20: return  $E_q$ 

```

Cross-document supersession extends the loop (line 8) to pairs (e_i, e_j) from *different* documents: if entity–object mentions overlap, $\tau_j > \tau_i$, and the NLI scorer detects semantic contradiction or entailment reversal, we install a cross-document SUPERSEDE edge. The *cross-source supersession* setting in Table 26 evaluates this in isolation.

D Controller-only diagnostic

Table 9 reports a controller-only diagnostic on 2,000 examples that bypasses the LLM. The diagnostic is framed as state/action evaluation rather than answer extraction: stale-action F1, conflict-action F1, supersession-edge F1, and contract-state accuracy.

Method	Stale-F1 $\uparrow$	Conf-F1 $\uparrow$	Edge-F1 $\uparrow$	Contract $\uparrow$
Vanilla + action	.120	.083	.052	.231
HoH-style + action	.471	.148	.103	.384
Self-reflect + action	.352	.213	.127	.341
VERA-LIFE controller	.932	.781	.743	.871

Table 9: Controller-only diagnostic on 2,000

VERA-LIFE-EVAL questions (no LLM generation). This isolates lifecycle state and action decisions from answer extraction. VERA-LIFE identifies stale, conflict, and supersession states before generation rather than relying on surface verbalisation. All baselines share the same JSON action schema.

E Contract objective derivation

Let $F_{\text{cov}}(E, q)$ denote only the coverage term in Equation 4. F_{cov} is monotone submodular under a cardinality constraint, so greedy selection attains the standard $(1 - 1/e)$ approximation (Nemhauser et al., 1978). The full objective subtracts redundancy and conflict penalties and is therefore not guaranteed to be monotone. We use it as a greedy heuristic with nonnegative-marginal stopping and report selector comparisons in Table 50.

F Full benchmark validation

Table 10 reports controller-level numbers on the full 8,500-question benchmark and four independently sampled 1,500-question subsets; the per-subset LifeEM standard deviation is .004.

Subset (seed)	LifeEM	SRR	CUR
Subset 1 (seed 42)	.623	1.000	.725
Subset 2 (seed 1337)	.618	.987	.712
Subset 3 (seed 2024)	.615	.993	.708
Subset 4 (seed 7)	.621	.993	.718
Mean $\pm$ Std	.619 $\pm$.004	.993 $\pm$.005	.716 $\pm$.008
Full 8,500	.611	.974	.694

Table 11: Multi-subset variance. Four independently sampled $n=1,500$ stratified subsets yield LifeEM std = .004, well below the gap between VERA-LIFE and the best baseline ($\Delta=.249$). The full-benchmark row is provided for reference; its slightly lower scores reflect the larger proportion of hard examples in the unstratified pool. Subset hashes are published with the artefact.

G REAL-mini construction and annotation

Table 12 details domain sources, inclusion criteria, and inter-annotator agreement; Table 13 adds a non-linear subset with multi-source supersession, retraction chains, and jurisdiction conflicts.

Domain	n	LifeEM gain	Edge F1	Src. attr.
API: deprec. + migr. + changelog	30	+.112	.683	.724
Legal: old + updated + jurisd.	30	+.098	.651	.697
Sci./med.: paper + guideline/retr.	30	+.087	.628	.672
News: report + corr. + follow-up	30	+.134	.712	.748
All non-linear	120	+.108	.672	.718
<i>Safe-action quality on 47 unresolvable cases</i>				
WARN/ABSTAIN judged reasonable				78%
Conflict detected but unresolved				23%

Table 13: Non-linear lifecycle subset of REAL-mini ($n=120$). Each domain contributes 30 cases with multi-source supersession, retraction/correction chains, or jurisdiction conflicts. LifeEM gains persist (+.108 average); conflict/supersession edge F1 (.672) and source attribution (.718) are lower than the linear subset, reflecting genuine difficulty. On 47 unresolvable cases, 78% of WARN/ABSTAIN actions are judged reasonable by human annotators, showing that the controller degrades gracefully rather than hallucinating resolutions.

H Hard-prefix conditioning ablation

Table 14 ablates the hard-prefix lifecycle conditioning; removing it drops LifeEM by .049, confirming prefix conditioning improves safe verbalisation without changing the model.

Configuration	LifeEM $\uparrow$	SRR $\uparrow$	Conf-F1 $\uparrow$	OAR $\downarrow$
VERA-LIFE (full)	.623	1.000	.720	.000
w/o hard prefix	.574	.982	.651	.012
soft instruction only	.551	.961	.618	.021
Action oracle + vanilla verb.	.487	.873	.542	.038
State estim. + shared verb.	.568	.978	.673	.008

Table 14: Hard-prefix conditioning ablation ($n=1,500$). Removing the hard prefix drops LifeEM by .049, confirming that prefix conditioning improves safe verbalisation. However, the state estimator with a shared verbaliser retains 73% of the gain over the best same-action baseline (.568 vs. .451), showing that lifecycle state estimation—not output formatting—is the primary contributor. The action-oracle row gives baselines ground-truth actions but a vanilla generator; the remaining gap (+.136) is explained by the lifecycle graph and contract selector.

I Conflict decomposition and answerability taxonomy

Table 15 decomposes conflict handling into detection, candidate grounding, attribution, and resolution; Table 16 groups conflicts by answerability so that resolvable conflicts are answered with newer-authority citation rather than uniformly refused.

Split	n	LifeEM $\uparrow$	SRR $\uparrow$	Sup-F1 $\uparrow$	CUR $\uparrow$	Conf-F1 $\uparrow$	OAR $\downarrow$
Current	1,500	.638	—	—	—	—	.005
Past	1,500	.237	—	—	—	—	.011
Stale-only	1,500	.961	.974	—	—	—	—
Update	2,000	.678	—	.523	.694	—	.004
Conflict	1,000	.568	—	—	—	.698	.009
Multi-hop	1,000	.392	—	—	—	—	.006
Full 8,500	8,500	.611	.974	.523	.694	.698	.008
1,500 subset	1,500	.623	1.000	.546	.725	.720	.000

Table 10: Scale validation on full VERA-LIFE-EVAL (8,500 questions, controller-level lifecycle metrics). Full-benchmark LifeEM (.611) is within .012 of the 1,500 generation subset (.623), confirming that the lifecycle conclusions are not subset artefacts. Minor differences reflect difficulty variation across the larger sample. Per-split SRR, Sup-F1, CUR, and Conf-F1 are reported where the metric’s denominator matches the split.

Panel A: Conflict explanation decomposition (conflict split, $n=191$)			
Submetric	VERA-LIFE	HoH+act	Vanilla+act
Detection F1 (trigger fires)	.823	.412	.187
Candidate grounding F1	.748	.321	.142
Both candidates correct	.672	.218	.097
Source attribution acc.	.761	.287	.108
Resolution appropriateness	.814	.523	.312
Hallucinated candidates	5.2%	18.7%	31.4%
Panel B: Constrained span-grounded extractor improvement			
Submetric	+Extractor	Original	Δ
Candidate grounding F1	.812	.748	+0.064
Both candidates correct	.738	.672	+0.066
Source attribution acc.	.803	.761	+0.042
Hallucinated candidates	2.8%	5.2%	-2.4
Panel C: Human evaluation ($n=200$ conflict cases, 3 annotators)			
Criterion	VERA-LIFE	HoH+act	Fleiss κ
Clearly explains conflict	78%	41%	.76
Grounded in evidence	82%	38%	.79
Avoids false single-value assertion	89%	52%	.84
Has source/time attribution	73%	27%	.73
User would trust	68%	31%	.71
Preference vs baseline	71%	—	.78

Table 15: Conflict explanation audit. **Panel A:** decomposing conflict performance into detection, candidate grounding, both-candidates-correct, source attribution, and resolution. Detection is already strong (.823); the bottleneck is candidate grounding and source attribution. **Panel B:** adding a constrained span-grounded extractor that requires candidate spans to be locatable in evidence improves grounding F1 by +.064 and reduces hallucinated candidates to 2.8%. **Panel C:** human evaluation of 200 conflict cases shows VERA-LIFE explanations are preferred 71% of the time because they avoid unsupported single-value assertions and expose evidence sources. Fleiss $\kappa \geq .71$ across all criteria.

Conflict type (n)	Correct action	VERA-LIFE	Baseline
True unresolved (87)	EXPLAIN	.851	.412
Resolvable by newer (48)	ANSWER + cite	.781	.542
Jurisdiction/scope (31)	qualified answer	.726	.387
Non-conflict paraphrase (25)	ANSWER	.912	.680
Overall (191)	—	.823	.487

Table 16: Post-hoc conflict answerability taxonomy on the 191-item generation subset; cells are action accuracy. This diagnostic does *not* change the official Conflict-split gold: LifeEM still credits EXPLAIN_CONFLICT (or an explicit conflict mark) as in section 6. Resolvable, scoped, and paraphrase cases are reported here to show that a finer action gold is possible in future work; the largest gap remains on true unresolved conflicts, where baselines tend to force a single answer.

J False warning and provenance analysis

Table 17 reports false-warning, false-conflict, and over-abstention rates on current-valid evidence, together with domain-level appropriateness and the provenance hierarchy used for tie-breaking.

Domain	Source types	n	Linear	Non-linear	κ	Lifecycle label source
Product / API	Official docs, changelogs, issue trackers, migration guides	180	150	30	.83	Deprecation notices, version diffs, changelog dates
Policy / legal	Statutes, regulations, court decisions, agency notices	160	130	30	.79	Effective dates, amendment markers, jurisdiction notes
Scientific / medical	Guidelines, meta-analyses, peer-reviewed trials, preprints	140	110	30	.81	Retraction notices, guideline updates, erratum dates
News corrections	Original reports, corrections, retractions, follow-up articles	170	140	30	.84	Correction timestamps, editor notes, retraction notices
Total / Avg.		650	530	120	.82	

Table 12: REAL-mini construction details. Each domain combines linear revision chains and non-linear multi-source cases. Lifecycle labels are derived from domain-specific markers (e.g., deprecation notices for API, retraction notices for scientific). Three annotators independently labelled lifecycle states; inter-annotator agreement (Fleiss κ) ranges from .79 (policy/legal, due to jurisdiction ambiguity) to .84 (news, where correction markers are unambiguous). Overall κ = .82 exceeds the .75 threshold for substantial agreement. Disagreement cases are adjudicated by a fourth annotator; the primary disagreement source is boundary ambiguity in validity intervals for policy and scientific domains.

Panel A: False warning / false conflict rates			
Diagnostic	VERA-LIFE	HoH+act	Vanilla+act
FWR-current (WARN on current)	3.4%	8.7%	1.2%
FCR (CONFLICT on non-conflict)	6.1%	4.3%	2.8%
OAR (abstain on answerable)	0.0%	6.5%	14.5%
Panel B: Warning appropriateness (REAL-mini)			
Domain	Appropriate%	FWR	FCR
Product / API docs	84%	2.8%	5.1%
Policy / legal updates	79%	4.2%	7.3%
Scientific / medical	86%	2.1%	4.8%
News corrections	78%	3.9%	6.7%
Average	81%	3.2%	5.9%
Panel C: Domain-specific provenance hierarchy			
Domain	Prov. Δ LifeEM	Src. attr.	Δ FWR
Product / API docs	+0.042	.812	-.8
Policy / legal	+0.068	.781	-1.2
Scientific / medical	+0.053	.768	-.6
News corrections	+0.037	.793	-.9

Table 17: False warning and provenance analysis.

Panel A: FWR-current (3.4%) and FCR (6.1%) quantify the cost of lifecycle-aware actions on current/non-conflict cases. VERA-LIFE’s false warning rate is low despite aggressive stale rejection; vanilla+action over-abstains (14.5%). **Panel B:** domain-level human judgement on REAL-mini finds 81% of warnings appropriate; scientific/medical shows the highest appropriateness (86%) reflecting cautious behaviour in high-risk domains. **Panel C:** domain-specific provenance scoring (e.g., guideline > preprint for medical, statute > blog for legal) improves LifeEM by +0.037+0.068 and reduces false warnings.

K Action-conditioned calibration

Table 18 reports action-conditioned binned ECE (LCE) and risk-coverage AUC; baselines suffer catastrophic stale/conflict LCE (.841/.793), while VERA-LIFE stays calibrated under every action.

Panel A: Action-conditioned LCE				
Configuration	ans ↓	stale ↓	conf ↓	all ↓
VERA-LIFE (full)	.182	.089	.127	.267
VERA-LIFE w/o policy	.073	.841	.793	.073
HoH-style + action	.387	.612	.583	.542
Vanilla RAG + action	.518	.892	.847	.685
Panel B: Risk-coverage analysis				
Configuration	RC-AUC ↑	Safe-act. ↑	Coverage ↑	
VERA-LIFE (full)	.847	.912	.854	
VERA-LIFE w/o policy	.712	.487	1.000	
HoH-style + action	.673	.612	.813	
Vanilla RAG + action	.548	.378	.871	

Table 18: Action-conditioned calibration and risk-coverage analysis. Panel A reports LCE conditioned on the emitted action. **Panel A:** the w/o-policy ablation achieves the lowest overall LCE (.073) because it selects ANSWER on 100% of items, yielding a narrow confidence distribution with low binned error—but its stale-conditional (.841) and conflict-conditional (.793) LCE are catastrophic. VERA-LIFE’s higher overall LCE reflects a richer action distribution; its stale (.089) and conflict (.127) calibration are the best in the table. **Panel B:** risk-coverage AUC confirms that VERA-LIFE achieves the best trade-off between safe actions and coverage. The w/o-policy ablation has full coverage but only 48.7% safe-action rate.

L Component-swap robustness

Table 19 swaps each VERA-LIFE component for a strong same-action baseline; the relative ranking is preserved and all swaps keep VERA-LIFE $\geq +.144$ LifeEM ahead of HoH-style.

Swapped component	LifeEM $\uparrow$	Δ	SRR $\uparrow$	Ranking
VERA-LIFE (default)	.623	—	1.000	#1
NLI: DeBERTa $\rightarrow$ RoBERTa-lg	.607	-.016	.987	#1
Reranker: BGE $\rightarrow$ MiniLM-L6	.614	-.009	.993	#1
Generator: Qwen3 $\rightarrow$ Llama-3.1-8B	.591	-.032	.987	#1
Timestamp: regex $\rightarrow$ Qwen3-0.6B	.618	-.005	1.000	#1
θ_p : .10 $\rightarrow$.05	.619	-.004	1.000	#1
θ_p : .10 $\rightarrow$.20	.608	-.015	.993	#1
All swaps combined	.573	-.050	.974	#1
Best baseline (HoH+act)	.428	—	.487	—

Table 19: Component-swap robustness ($n=1,500$). Each row replaces one component with an alternative; the ranking column shows whether VERA-LIFE still outperforms all baselines. Individual swaps cost $\leq .032$ LifeEM; even swapping all components simultaneously (.573) remains $+ .145$ above the best baseline. The largest sensitivity is to the generator ($-.032$), the smallest to the timestamp parser ($-.005$) and provenance threshold θ_p ($-.004$ to $-.015$). System ranking is stable across all configurations.

M Error propagation analysis

Table 20 sweeps gold-noise injection at 10%, 20%, and 30% on the four lifecycle variables; degradation is smooth and sub-linear, with no catastrophic propagation.

Perturbation	Δ LifeEM	Δ SRR	Primary effect
Supersession edge $\pm 10\%$ noise	-.028	-.018	Update/stale
Validity interval $\pm 20\%$ IoU	-.039	-.007	Past/update
Provenance $\theta_p \pm 0.1$	-.012	-.004	False stale
NLI conflict score $\pm 15\%$	-.021	-.003	Conflict
Timestamp ± 1 day jitter	-.008	-.002	Boundary
Combined (all perturbations)	-.051	-.023	All splits

Perturbed VERA-LIFE LifeEM: .572; best baseline: .428

Table 20: Error propagation under estimator noise ($n=1,500$). Each row perturbs one lifecycle estimator by the indicated noise level and measures downstream LifeEM impact. Validity-interval noise has the largest effect ($-.039$) because it directly influences past/update split decisions. Even under combined noise, VERA-LIFE (.572) remains $+ .144$ above the best baseline (.428), showing that the ranking is robust to realistic estimator error cascades. Supersession edges and NLI scores interact: joint perturbation (.051) is less than the sum of individual effects (.049), suggesting partial redundancy in the lifecycle graph.

N Cross-dataset transfer details

Table 21 extends the main cross-dataset table with per-task breakdowns on HoH-QAs, StreamingQA, and TempReason; all transfer gains are positive and align with the main benchmark.

Method	FRAMES temporal	HoH-QAs stale-only		HoH-QAs update	
	AnsF1	LifeEM	EM	AnsF1	EM
Closed-book	.050	.000	.000	.013	.000
Vanilla RAG	.111	.010	.020	.241	.370
HoH-style	.110	.275	.020	.368	.495
VERA-LIFE	.054	.995	.020	.661	.680

Table 21: Cross-dataset transfer. On *FRAMES temporal* (Krishna et al., 2025) ($n=278$) the title-only evidence is too weak to support a definite answer, so VERA-LIFE abstains 84.5% of the time and trades AnsF1 for safety. On *HoH-QAs stale-only* ($n=200$, outdated revision is the only evidence) VERA-LIFE reaches LifeEM .995 versus .275 for HoH-style stale-aware prompting (a $3.6\times$ gain). On *HoH-QAs update* ($n=200$, mixed outdated and current revisions) VERA-LIFE reaches EM .680 versus .495 for HoH-style and .370 for vanilla RAG. All systems share the same Qwen3-8B generator and prompt family; only the controller logic differs.

O Cost-quality, significance, case studies, and end-to-end breakdown

Table 22 relates retained quality (LifeEM) to controller latency and prompt budget; Table 23 reports $1,000\times$ bootstrap 95% CIs for every main-table cell; Table 24 compiles four full traces with ledger evidence and action audits; Table 25 breaks down the end-to-end latency budget ($+42\%$, $\geq 92\%$ retention).

Method	sec/q	tok/q	LifeEM	SU	SU/sec
Closed-book	0.08 (0.56 $\times$)	17	.071	-0.36	-4.4
Vanilla RAG	0.15 (1.00 $\times$)	17	.217	+0.14	+0.9
Long-context (20 ev)	0.15 (1.02 $\times$)	17	.217	+0.15	+1.0
HoH-style	0.14 (0.95 $\times$)	17	.374	+0.34	+2.4
VERA-LIFE	0.14 (0.93 $\times$)	17	.623	+1.51	+10.9

Table 22: Generation-only cost-quality Pareto on 1,500 VERA-LIFE-EVAL questions, batch 6, greedy decoding (max 24 new tokens), single RTX 4090D. sec/q is *generation* latency after retrieval and controller setup; it is *not* end-to-end wall-clock. Under this generation-only view VERA-LIFE is $0.93\times$ vanilla RAG and $4.5\times$ the SafeUtility of HoH-style prompting. End-to-end latency, including graph, NLI, and reranking, is $+42.4\%$ over vanilla RAG (Table 25).

Method	AnsF1 (95% CI)	LifeEM (95% CI)
Closed-book Qwen3-8B	.067 [.054, .080]	.071 [.057, .085]
Vanilla RAG (Hybrid)	.260 [.241, .279]	.217 [.196, .238]
Long-context RAG (20 ev)	.259 [.240, .278]	.217 [.196, .238]
HoH-style stale-aware	.339 [.320, .358]	.374 [.351, .397]
VERA-LIFE (full)	.402 [.382, .422]	.623 [.597, .649]

Table 23: Bootstrap 1,000× resample 95% confidence intervals on $n=1,500$ VERA-LIFE-EVAL questions. CI centres match the main table exactly (same frozen prediction dump). All gaps between VERA-LIFE and every baseline are non-overlapping, indicating significance at $p < .05$.

P Construction-metadata isolation

Table 26 hides gold-private fields, public metadata, lifecycle markers, and cross-source supersession in turn; VERA-LIFE remains above the best same-action baseline in every regime, ruling out construction-metadata leakage. Table 27 and Table 31 extend the test with decoy-resilience, ledger-forcing, shortcut probes, adversarial corruption, and input-field parity.

Setting	LifeEM ↑	SRR ↑	CUR ↑	OAR ↓
Full metadata (upper bound)	.623	1.000	.725	.000
Gold-private hidden	.612	.987	.718	.005
Public metadata only	.591	.968	.697	.012
Text-only marker-removed	.483	.821	.581	.042
Cross-source supersession	.572	.945	.668	.021
Best same-action baseline	.451	.394	.478	.082

Table 26: Construction-metadata isolation ($n=1,500$). *Gold-private hidden*: all metadata used to construct gold answers is withheld from the system. *Public metadata only*: only deployable timestamps, deprecation notices, and version tags are provided. *Text-only marker-removed*: explicit lifecycle markers (“deprecated”, “superseded”, “updated”, “correction”) are removed; inference relies on textual disagreement and NLI signals. *Cross-source supersession*: old and new evidence come from different domains or sites. VERA-LIFE remains above the best same-action baseline in every setting, confirming that gains are not artefacts of gold-construction metadata leakage.

Decoy type (n)	VERA-LIFE acc.	Baseline acc.	Δ
New timestamp, irrelevant (80)	.912	.623	+.289
Old timestamp, still valid (80)	.863	.578	+.285
“Updated” non-entity change (60)	.883	.542	+.341
Partial-scope correction (60)	.817	.487	+.330
Jurisdiction/scope, not conflict (40)	.825	.512	+.313
Avg. ($n=320$)	.864	.553	+.311

Table 27: Decoy / adversarial negative control ($n=320$). We construct decoy evidence designed to trigger metadata-shortcut heuristics. VERA-LIFE correctly ignores temporally irrelevant evidence (91.2%), does not reject old-but-valid evidence (86.3%), and distinguishes genuine entity updates from surface lexical changes (88.3%). These results rule out simple metadata-matching explanations for the lifecycle gains.

Method	LifeEM ↑	SRR ↑	Δ vs orig.
Self-reflect + ledger template	.492	.523	+.041
ReAct agentic + ledger template	.478	.498	+.041
HoH-style + ledger template	.465	.512	+.037
VERA-LIFE full	.623	1.000	—

Table 28: Baseline ledger-forcing control ($n=1,500$). All baselines receive the same lifecycle ledger JSON template as VERA-LIFE. Ledger templates improve baselines by $+.037$ – $+.041$ LifeEM, confirming that structured output helps; however, VERA-LIFE remains $+.131$ above the best ledger-equipped baseline, showing that the lifecycle state estimation—not the output schema—drives the primary advantage.

Shortcut probe (text hidden)	LifeEM ↑	SRR ↑	CUR ↑	OAR ↓
Timestamp-only heuristic	.312	.447	.328	.183
Role-label-only	.298	.422	.315	.197
Source-title-only	.274	.398	.291	.218
Evidence-order-only	.261	.375	.278	.234
Marker-only (string match)	.346	.482	.362	.161
Strongest probe (marker-only)	.346	.482	.362	.161
<i>Marker-only stress: lifecycle markers removed from text</i>				
Marker-only after removal	.197	.183	.172	.305
VERA-LIFE marker-removed	.483	.821	.581	.042
Δ over best same-action baseline	+.082	+.427	+.103	−.040

Table 29: Artifact-shortcut probes ($n=1,500$, full benchmark). Each probe receives a single metadata field with the textual evidence hidden, isolating the contribution of any individual construction artefact. No probe reaches LifeEM .35, while VERA-LIFE retains .483 LifeEM after explicit lifecycle markers are removed from the text, $+.082$ above the best same-action baseline. After marker removal the marker-only string-match probe collapses to near-random (.197), but VERA-LIFE’s gain persists because its lifecycle estimation does not reduce to surface marker matching.

Stale-verify	<i>As of today, is the answer to “Who are the three workers in the simple example” still “Alice, Bob, and Dora”?</i>
Gold	<i>no</i>
Vanilla RAG	answer :: Yes, the answer is still Alice, Bob, and Dora.
HoH-style	answer :: Alice, Bob, and Dora.
VERA-LIFE	warn_stale :: no longer current; was Alice, Bob, and Dora
Update	<i>What is Ian Veneracion’s birth year?</i>
Gold	<i>1975</i>
Vanilla RAG	answer :: 1973
HoH-style	answer :: 1973
VERA-LIFE	answer :: 1975
Conflict (honest failure)	<i>What percentage of the population in the Davao Region are Muslims?</i>
Gold	<i>3.55%</i>
Vanilla RAG	answer :: 27% of the population are Muslims.
HoH-style	answer :: no longer current
VERA-LIFE	explain_conflict :: conflicting 63% / 99.32%

Table 24: Three representative cases. On *stale-verify* VERA-LIFE emits an explicit WARN_STALE prefix and names the superseded value, where vanilla and HoH-style prompting answer with the obsolete value verbatim or implicitly. On *update*, the contract selector promotes the current-revision evidence so the generator returns the new value 1975. The *conflict* row is shown deliberately as a failure mode: the policy correctly fires EXPLAIN_CONFLICT, but the two surfaced candidates (63% and 99.32%) come from different paragraphs of an outdated revision, neither matching the gold answer; this is the dominant residual error pattern on the conflict split.

Corruption	VERA-LIFE	LifeEM	Δ vs clean	Best base. Δ
Clean (reference)	.623	—	—	—
Timestamp shuffle	.587	—	−.036	−.218
Source-title masking	.596	—	−.027	−.171
Evidence-order shuffle	.604	—	−.019	−.182
Role-label swap	.342	—	−.281	−.058
Marker removal	.483	—	−.140	−.094
Avg. corruption drop	−.101	—	—	−.145

Table 30: Adversarial corruption stress ($n=1,500$). Each row applies one targeted corruption. VERA-LIFE is robust to timestamp shuffling and order shuffling (−.036, −.019), confirming it does not depend on order or absolute timestamp. Role-label swap (−.281) is the only large drop, and is symmetric across baselines, showing that role information is read but never trusted in isolation (see Table 46). Average drop (−.101) is substantially smaller than baselines’ (−.145), so adversarial corruption widens rather than closes the gap.

Q REAL-blind transfer

Table 32 and Table 33 report REAL-blind ($n=1,200$) and REAL-blind++ ($n=1,000$) with source-disjoint splits; Table 34 gives four full per-domain traces and Table 35 releases the REAL-mini datasheet.

Domain	LifeEM gain $\uparrow$	SRR $\uparrow$	OAR $\downarrow$	Conf-F1 $\uparrow$
Product / API docs ($n=320$)	+ .138	.835	.072	.723
Policy / legal ($n=280$)	+ .121	.803	.088	.692
Scientific / medical ($n=260$)	+ .098	.776	.095	.651
News corrections ($n=340$)	+ .152	.851	.065	.748
Avg. ($n=1,200$)	+ .128	.817	.079	.705
<i>Metadata regime ablation (avg. across domains)</i>				
Timestamp-only	+ .109	.793	.085	.672
Text-only	+ .068	.712	.098	.581
Marker-removed	+ .112	.798	.082	.683

Table 32: REAL-blind cross-domain transfer ($n=1,200$). Unlike REAL-mini, REAL-blind uses pre-defined source lists and random sampling within fixed time windows; no author selection is involved. Gold annotations are human-labelled with public source provenance; no Wikipedia revision metadata is used. LifeEM gain is over the best same-action baseline per domain. All domains show positive gains ($\geq +.098$); marker removal retains 87.5% of the gain. The lifecycle contract is not a Wikipedia/HoH artefact.

R Text-only deployment analysis

Table 36, Table 37, and Table 38 report the text-only deployment regime in which all systems are denied structured metadata: VERA-LIFE retains +.062 LifeEM over the best same-action baseline and the dominant residual error is the missing-lifecycle-cue (32%).

Component	VERA-LIFE ms/q	Vanilla RAG	Long-ctx RAG	GPU MiB
Query parse + temporal intent	7	0	0	32
Multi-snapshot retrieval	22	18	18	1,820
Reranker (Qwen3-Reranker-0.6B)	19	16	16	1,210
NLI + lexical edge inference	26	—	—	1,340
Lifecycle graph construction	6	—	—	11
Evidence-contract selector	14	—	—	9
Risk-calibrated action policy	4	—	—	6
Ledger serialisation	9	—	—	14
Generation (Qwen3-8B, 24 tok)	138	138	154	14,820
Total	245	172	188	19,262
Overhead vs vanilla	+42.4%	1.00×	1.09×	—
Vs full-context oracle (all snapshots)	—	—	—	—
with delta-compressed snapshots	—	—	412	31,540
VERA-LIFE ledger replaces full ctx	1.68× cheaper	—	—	—
<i>Quality-retention sweep</i>				
VERA-LIFE full	LifeEM .623	—	—	—
small NLI (DeBERTa-base)	.604 (95.5% retained)	—	—	—
no reranker	.591 (92.2% retained)	—	—	—
no NLI, lexical edges only	.563 (87.5% retained)	—	—	—

Table 25: End-to-end cost breakdown on a single RTX 4090D (batch 6, greedy decoding, 24 new tokens, $n=1,500$). Generation-only latency is $0.93\times$ vanilla (Table 22); the +42.4% figure here is *end-to-end* wall-clock, including query parse, retrieval, reranking, NLI, graph construction, and ledger serialisation. The two numbers measure different scopes and are not in conflict. VERA-LIFE remains $1.68\times$ cheaper than long-context retrieval over all evidence snapshots. Replacing NLI with DeBERTa-base or removing the reranker still retains $\geq 92\%$ of the LifeEM gain. The controller is training-light and modular, not free at inference.

Method (text-only)	LifeEM $\uparrow$	SRR $\uparrow$	CUR $\uparrow$	OAR $\downarrow$	Conf-F1 $\uparrow$
Vanilla RAG + action	.387	.198	.412	.152	.108
HoH-style + action	.412	.321	.448	.118	.195
Self-reflect + action	.442	.348	.467	.095	.258
ReAct agentic + action	.427	.332	.453	.102	.231
GraphRAG + action	.398	.278	.428	.128	.168
LightRAG + action	.408	.298	.438	.115	.182
VERA-LIFE text-only	.504	.718	.583	.052	.512

Table 36: Text-only comparison: all systems receive text evidence with no explicit lifecycle metadata ($n=1,500$). VERA-LIFE text-only outperforms the best same-action baseline (Self-reflect) by +.062 LifeEM (bootstrap 95% CI [+0.038, +.086], $p<.01$). The gap is modest compared to full metadata (+.172) but statistically meaningful across all splits. Largest per-split gains appear on *update* (+.083) and *stale-only* (+.072), where textual disagreement signals partially compensate for missing timestamps.

Text-only variant	LifeEM $\uparrow$	Δ vs nat.	Δ vs base.
Natural-marker retained	.504	—	+.062
Marker-removed	.483	−.021	+.041
+ timestamp removed	.476	−.028	+.034
+ order shuffled	.461	−.043	+.019
<i>Reference (text hidden)</i>			
Role-label-only probe	.298	—	−.144
Marker-only probe	.346	—	−.096
Best same-action baseline	.442	—	—

Table 37: Text-only regime decomposed into four variants ($n=1,500$). Even after explicit lifecycle markers are removed, VERA-LIFE retains .483 LifeEM and stays +.041 above the best same-action baseline. Further removing timestamps and shuffling evidence order costs another .022 but the system remains positive (+.019). The drop from removing markers (−.021) is smaller than the drop the role-label-only and marker-only probes suffer when they lose their single signal ($\rightarrow$.298, .346), showing that VERA-LIFE’s lifecycle estimation aggregates signals rather than reading a single artefact.

System / regime	TS	ST	RID	RL	DM	NM	OR	PR
Same-action baselines (full-metadata regime)								
Vanilla RAG + action	✓	✓	✗	✗	✓	✓	✓	✗
HoH-style + action	✓	✓	✗	✓	✓	✓	✓	✗
Self-reflect + action	✓	✓	✗	✓	✓	✓	✓	✗
ReAct / GraphRAG / LightRAG + action	✓	✓	✗	✓	✓	✓	✓	✗
VERA-LIFE full	✓	✓	✗	✓	✓	✓	✓	✓
VERA-LIFE under deployment regimes								
timestamp-only	✓	✓	✗	✗	✗	✓	✓	✗
text-only natural	✗	✓	✗	✗	✗	✓	✓	✗
text-only marker-removed	✗	✓	✗	✗	✗	✗	✓	✗
text-only + order shuffled	✗	✓	✗	✗	✗	✗	✗	✗
gold-private hidden	✓	✓	✗	✓	✓	✓	✓	✗

Private construction-only (never seen at inference): revision IDs, deletion logs, edit summaries, gold intervals.

Table 31: Input-field parity across baselines and VERA-LIFE regimes. Column codes: TS = timestamp, ST = source title, RID = revision ID, RL = role label, DM = deprecation marker, NM = natural marker, OR = evidence order, PR = provenance. Same-action baselines receive an identical serialisation and the identical five-way JSON action schema; the only VERA-LIFE-exclusive input is the optional structured *provenance* field, which is also evaluated under removal. Wikipedia revision IDs, deletion logs, edit summaries, and gold validity intervals are construction metadata and are *never* given to any system at inference time. All reported gains hold under this parity.

Error type	% of errors	Description
Missing lifecycle cue	32%	Text lacks update signals
Lexical update missed	24%	Cue present but NLI/graph missed
Conflict over-trigger	15%	Scope diff. → false contradict.
Stale under-trigger	18%	Old evidence treated as valid
Generator assertion err.	11%	Ledger correct, verbalis. wrong

Table 38: Text-only error decomposition ($n=342$ errors on text-only prediction). The dominant failure mode (32%) is *missing lifecycle cue*: the text contains no lexical or structural signal of update, so neither graph construction nor NLI can detect the lifecycle transition. This establishes an information-theoretic limit on pure-text lifecycle inference and motivates minimal deployable metadata (e.g., timestamps alone recover most of the gap; see Table 2).

S Past-split diagnostic

Table 39 attributes past-split failure to retrieval (34%), selector (28%), action (22%), and generation (16%); the target-time-aware selector lifts past LifeEM .142 → .328.

Past variant	AnsF1 ↑	LifeEM ↑
Original selector (current-biased)	.103	.142
Target-time-aware selector	.198	.247
+ date-expanded retrieval	.253	.328
Oracle retrieval + original selector	.178	.231
Original retrieval + oracle selector	.215	.278
Oracle ledger + generator	.312	.408
Vanilla RAG (reference)	.217	.152

Table 39: Past-split diagnostic ($n=267$). The original system had a residual current-revision bias in the selector, which penalised valid historical evidence. Correcting the validity term to be explicitly target-time aware improves AnsF1 by +.095; adding date-expanded retrieval further lifts AnsF1 by +.055. Oracle variants identify retrieval (34% of failures) and selector (28%) as the dominant bottlenecks, not lifecycle definition or generation. With both fixes, the past-split gap versus Vanilla RAG reverses from $-.114$ to $+.036$ AnsF1 while stale/update LifeEM changes by $\leq .005$.

T Expanded human utility evaluation

Table 40 and Table 41 report $n=800$ blind stratified human evaluation with six criteria and the Spearman/Kendall correlations between LifeEM and human judgements ($\rho=.73$).

Annotator protocol. All human evaluation studies in this paper (the $n=800$ blind utility audit, the $n=400$ expanded six-criterion audit, the $n=200$

Domain (n)	LifeEM gain	SRR	OAR	Marker-rem. retention	Cross-source super.
Product / API (250)	+142	.842	.071	88.1%	.604
Policy / legal (260)	+118	.812	.083	84.7%	.571
Scientific / medical (240)	+097	.781	.091	81.3%	.548
News correction (250)	+151	.859	.063	89.4%	.627
Avg. ($n=1,000$)	+127	.824	.077	85.9%	.588
Min. across domains	+097	.781	.063	81.3%	.548
<i>Audit independence</i>					
Sources	disjoint from VERA-LIFE-EVAL	—	—	—	—
Annotators	4 domain experts (1 per domain), 3-way for legal/medical	—	—	—	—
IAA κ	.82 (overall); .77 medical; .79 legal	—	—	—	—
Hard negatives (%)	18.4% (irrelevant new ts + old-but-valid)	—	—	—	—
False warning rate	medical 7.2%, legal 9.8%, news 5.3%, API 4.1%	—	—	—	—

Table 33: REAL-blind++ source-disjoint external evaluation ($n=1,000$, four domains, sources independent of VERA-LIFE-EVAL construction). All four domains show positive LifeEM gain ($\geq +.097$). Scientific/medical has the lowest gain but the highest safe-action rate, matching the prior that high-stakes domains require more conservative behaviour. Marker-removed retention is $\geq 81.3\%$ in every domain, and cross-source supersession LifeEM is $\geq .548$. These results are an independent reproduction of the lifecycle-faithfulness gain outside the Wikipedia-style construction pipeline.

conflict explanation audit, and the REAL-mini $\kappa=.82$ lifecycle labelling) follow a single protocol.

- **Annotator pool.** Three trained annotators per study with graduate-level English proficiency and prior NLP/QA annotation experience, recruited via the authors’ research network; no crowdsourcing platform was used. Annotators were not told which system produced which output (blind A/B/C labels) and were randomly assigned to stratified question batches.
- **Compensation.** Annotators were compensated at a rate calibrated to be at or above their local hourly research-assistant wage, based on a pre-study time estimate of 30–60 seconds per item; the per-item rate is the same across all studies.
- **Instructions** (verbatim template). “You will see a question, four pieces of retrieved evidence with timestamps, and the answer produced by an anonymous system. For each criterion, judge the answer independently of any other answer you have seen on the same question. (i) *Factual correctness*: is the asserted fact consistent with the most recent valid evidence? (ii) *Conflict awareness*: if evidence disagrees, does the answer surface the disagreement (yes/partial/no)? (iii) *Action appropriateness*: is the action label (ANSWER/WARN_STALE/EXPLAIN_CONFLICT/RETRIEVE_MORE/ AB-

STAIN) appropriate? (iv) *Source attribution*: is the cited evidence the right piece for the asserted claim? (v) *User trust*: would you trust this answer in a deployed product? (vi) *Harmful assertion avoidance*: does the answer avoid stating an outdated or unsupported value as current? Score each criterion on $\{0, 0.5, 1\}$ and provide a one-sentence rationale.”

- **Quality control.** 10% of items are duplicated as gold-checks with a pre-defined expected label; annotators with gold-check accuracy below 0.8 are retrained on a held-out batch before their judgements enter the pool. Inter-annotator agreement (Fleiss κ) is reported per criterion in Table 40 ($\kappa \geq .74$).
- **Ethics.** No annotator interacts with end users or identifiable personal data; all evidence comes from publicly available encyclopaedic, product, policy, scientific, or news sources. The protocol was reviewed under the authors’ institutional guidelines for low-risk human evaluation studies.

Domain	Old → new evidence	Baseline failure	VERA-LIFE
API	<i>Boto3</i> v1.20: ContentMD5 accepted → v1.34: ContentMD5 deprecated, use ChecksumSHA256.	HoH cites v1.20, no deprecation note.	WARN_STALE; ledger marks v1.20 superseded; suggests ChecksumSHA256.
Legal	<i>State A 2020</i> : 8am–9pm → <i>2024 amend.</i> : 9am–8pm; State B unchanged.	Self-reflect cites the 2020 statute; no jurisdiction.	EXPLAIN_CONFLICT: State A 9–8pm (2024); State B 8–9pm; 2020 row superseded.
Medical	<i>2018</i> : drug <i>X</i> at d_1 first-line → <i>2023</i> : <i>X</i> first-line, max dose d_2 , renal cautions.	ReAct gives d_1 from 2018, no caution.	WARN_STALE on dose, ANSWER on drug; ledger separates valid (drug) from updated (dose).
News	<i>Original</i> : 14 casualties → <i>Next-day correction</i> : 9; later follow-up notes ongoing investigation.	GraphRAG gives “14” from the original article.	ANSWER 9; ledger cites correction; follow-up flagged as ongoing, not a second conflict.

Table 34: Four full cross-domain case studies (one per REAL-blind domain). Each row shows the evidence-time chain, a representative same-action baseline failure, and VERA-LIFE’s action plus ledger. Selected from the audit ($n=1,200$, Table 32); full traces in the supplementary release.

Criterion	VERA-LIFE	Self-refl.	HoH	κ
Factual correctness	.78	.62	.58	.81
Conflict awareness	.85	.41	.38	.79
Action appropriateness	.82	.53	.51	.83
Source attribution	.74	.32	.29	.76
User trust	.71	.38	.34	.74
Harmful assertion avoidance	.89	.54	.51	.86
Overall preference	72%	17%	11%	.78

Table 40: Expanded human utility evaluation ($n=400$, 3 annotators). Six evaluation dimensions are scored independently. VERA-LIFE’s largest advantages are on harmful-assertion avoidance (.89 vs. .54) and conflict awareness (.85 vs. .41), confirming that the lifecycle controller’s value lies in preventing unsafe assertions rather than only improving answer accuracy. Fleiss $\kappa \geq .74$ across all criteria.

U Metric decomposition and threshold sensitivity

Table 42 decomposes the LifeEM gain across six independently-scored metric families; Table 43, Table 44, and Table 45 report IoU/threshold sensitivity and the ambiguous-interval taxonomy with stratified human audit.

Metric family	VERA-LIFE	Best SA	Gap	Ledger-free?
Answer-only EM/F1	.402	.375	+.027	✓
Valid-evidence F1	.712	.587	+.125	partial
State accuracy	.867	.623	+.244	✓ [†]
Action accuracy	.840	.540	+.300	✓ [†]
Human pref. (400)	72%	28%	+44%	✓
LifeEM (action-cond.)	.623	.451	+.172	✗
Unsafe stale assertion	.042	.187	−.145	✓

Table 42: Metric decomposition ($n=1,500$). We report VERA-LIFE against the strongest same-action baseline (Best SA = Self-reflect) across six metric families to audit whether the advantage is confined to the lifecycle-specific LifeEM. *Ledger-free?*: metric can be computed without VERA-LIFE’s structured ledger. [†]State and action accuracy require all systems to output state/action, which the same-action schema provides. VERA-LIFE gains are largest on lifecycle-specific dimensions (state, action, valid-evidence), modest on answer-only EM/F1, and strongly positive on human preference and unsafe assertion reduction—confirming that the advantage is not merely a metric-format artefact.

Datasheet item	Description
Source inclusion criteria	Public, freely accessible sources with verifiable lifecycle transitions; at least two temporally distinct versions of the same claim available
Date range	Evidence published between Jan 2018 and Dec 2025
Sampling frame	Pre-defined source lists per domain; random sampling within fixed time windows
Exclusion criteria	Non-English sources; claims with <2 evidence versions; ambiguous lifecycle state
Candidates before filtering	2,847 initial candidates $\rightarrow$ 650 after filtering
Annotator qualifications	Graduate-level researchers with domain expertise (1 per domain + 1 generalist)
Annotation interface	Custom web interface with evidence display, lifecycle-state dropdown, free-text justification
Adjudication process	3 independent annotations; disagreements resolved by 4th senior annotator
IAA by label type	Lifecycle state $\kappa=.82$; validity interval $\kappa=.76$; action $\kappa=.85$
Source-level split	Eval sources do not overlap with any dev/tune sources; vendors, outlets, journals are disjoint
Hard negatives included	$n=98$: old-but-valid (32), updated-but-claim-unchanged (28), same-entity-different-jurisdiction (22), retraction-unrelated-to-claim (16)
Hard-negative false-warning rate	$\leq 6.1\%$
Released artefacts	Source URLs, document IDs, annotation guidelines, discarded examples with reasons

Table 35: REAL-mini datasheet. The dataset follows Datasheets for Datasets (Geburu et al., 2021) principles. Source-level splitting ensures that eval evidence comes from unseen vendors, outlets, and journals. Hard negatives ($n=98$) test whether the system over-warns on lifecycle-neutral evidence.

IoU threshold	VERA-LIFE	Best baseline	Gap	Rank
$\tau=.3$ (lenient)	.648	.478	+170	1st
$\tau=.5$ (default)	.623	.451	+172	1st
$\tau=.7$ (strict)	.591	.417	+174	1st
Exact interval match	.571	.421	+150	1st
Kendall τ rank correlation across thresholds				.93

Table 43: Optional stricter LifeEM+IoU diagnostic ($n=1,500$). The headline LifeEM in the main tables does *not* conjunct interval IoU; this table additionally requires $\text{IoU} \geq \tau$ on ANSWER splits. Numbers at $\tau \in \{.3, .5, .7\}$ are identical to Table 44 and Table 64. Rankings are unchanged (Kendall $\tau=.93$). The exact-interval row is a stricter probe, not part of the shared three-threshold sweep.

System	LifeEM @.3	@.5	@.7	Rank τ
Vanilla RAG	.241	.218	.192	.94
Long-context (Gemma-3)	.252	.226	.197	.93
HoH-style	.398	.371	.338	.94
Self-reflect + action	.478	.451	.417	.93
ReAct + action	.462	.437	.402	.93
GraphRAG + action	.411	.389	.355	.92
LightRAG + action	.438	.412	.378	.93
VERA-LIFE full	.648	.623	.591	.95
Avg. gap to best base.	+170	+172	+174	—

Table 44: Optional LifeEM+IoU rank stability across $\{.3, .5, .7\}$ ($n=1,500$). The default @.5 column equals the reported action-conditioned LifeEM because the additional IoU filter is inactive at the default operating point used in the main tables. The @.3 and @.7 columns add an interval-overlap constraint on ANSWER splits and match Table 43 and Table 64. Long-context RAG uses Gemma-3-12B-IT; all other systems use Qwen3-8B. VERA-LIFE ranks first under all three thresholds (system-level Kendall $\tau=.93$).

V Isolation experiments and audits

The remaining isolation suite tables report the full same-action grid (Table 47), unified evaluation hashes, operational audits, selector comparisons, ledger decoding (Table 51), the metric card (Table 52), SafeUtility sensitivity, the selector budget sweep (Table 54), marker-removal ablations, and the target-time validity revision together with its conflict-grounding panel (Table 55). Table 57 adds a stratified human audit of VERA-LIFE-EVAL gold ($n=1,200$) that puts overall gold-answer noise at 5.6%. All result cells derive from one prediction dump and one metric script.

Human criterion	vs LifeEM ρ	τ	vs AnsF1 ρ	vs SafeUtility ρ
Factual correctness	.58	.49	.71	.54
Action appropriateness	.73	.61	.28	.76
Harmful-assertion avoidance	.81	.68	.14	.79
User helpfulness / task success	.67	.55	.42	.69
Conflict awareness	.76	.63	.19	.74
Source attribution	.62	.51	.31	.65
User trust	.69	.57	.35	.71
Aggregate (4-criterion avg.)	.69	.58	.31	.72
Overall human preference (binary)	.72	.60	.26	.74
<i>Stratified by lifecycle split</i>				
Current	.61	.51	.58	.63
Past	.65	.54	.39	.67
Stale-only	.82	.69	.12	.80
Conflict	.79	.66	.09	.77
Update	.76	.64	.23	.75
Multi-hop	.68	.57	.35	.70

Table 41: Spearman (ρ) and Kendall (τ) correlation between human utility judgements ($n=800$, 3 annotators) and three automatic metrics: LifeEM, AnsF1, and SafeUtility. LifeEM correlates strongly with action appropriateness ($\rho=.73$) and harmful-assertion avoidance ($\rho=.81$), while AnsF1 correlates best with factual correctness ($\rho=.71$) and weakly with safety (.14). This is the intended decomposition: LifeEM is an action-aware reliability metric that complements, rather than replaces, token-level answer F1.

Audit item	Final run field
Data split hash	sha256:a3f8. . . 7c21
Prediction dump hash	sha256:b5e2. . . 4d18
Metric script hash	sha256:c7a1. . . 9f35
Decoding config hash	sha256:d2b4. . . 6e72
Random seed list	[42, 1337, 2024, 7, 31]
Table 2 source run	run-20260418-unified
CI table source run	run-20260418-unified
Controller diag. source run	run-20260418-unified

Table 48: Unified evaluation audit. All reported tables originate from one frozen prediction dump and one metric script. Table 2, the bootstrap CI table, and the controller diagnostic share the same source run identifier; hashes are published with the artifact.

Selector	LifeEM $\uparrow$	Conf. err. $\downarrow$	Redund. $\downarrow$	Latency
Random selector	.413	.231	.167	8 ms
Coverage-only greedy	.587	.142	.085	12 ms
Penalised greedy (ours)	.623	.073	.041	14 ms
MMR-style selector	.601	.098	.052	18 ms
Small-set exhaustive oracle	.651	.048	.028	142 ms

Table 50: Selector comparison. The penalised greedy heuristic achieves 95.7% of the exhaustive-oracle LifeEM at 9.9% of its latency. Conflict and redundancy penalties improve LifeEM by +3.6 over coverage-only greedy at +2 ms.

Ledger audit	Compact ledger	Full JSON ledger
Max new tokens	24 answer tokens	256 ledger tokens
JSON parse success	n/a	97.8%
Claim coverage	.91	.96
State accuracy	.87	.92
Evidence-id validity	.84	.89
LifeEM vs no-ledger	+124	+137
Latency overhead	+0 ms	+38 ms

Table 51: Ledger and decoding audit. The compact ledger (24 tokens) provides 90.5% of the full-JSON LifeEM gain at zero additional latency. Full JSON achieves 97.8% parse success and higher claim coverage and state accuracy, at a 38 ms overhead.

Statistic	VERA-LIFE	Best baseline	2nd baseline
% settings ranked #1	91.7%	8.3% (HoH)	—
Kendall τ rank stability	.89	—	—
SafeUtility (min / med / max)	1.02 / 1.51 / 2.18	0.21 / 0.34 / 0.58	0.08 / 0.22 / 0.41

Table 53: SafeUtility sensitivity across 48 weight settings ($\lambda_s \in \{1, 2, 3, 5\}$, $\lambda_o \in \{0.25, 0.5, 1\}$, $r_w \in \{0.5, 1\}$, $r_c \in \{0.5, 1\}$). VERA-LIFE ranks first in 91.7% of settings; system rankings are stable (Kendall $\tau=.89$).

Ambiguous interval class (n)	IAA κ	Auto-gold acc.	VERA-LIFE LifeEM	Best baseline
Jurisdiction-specific legal update (80)	.73	.788	.618	.397
Weakened-not-retracted medical guideline (60)	.68	.742	.587	.362
News correction vs follow-up (70)	.76	.823	.652	.431
Old-but-valid historical answer (90)	.81	.867	.706	.484
Versioned-API deprecation overlap (50)	.79	.852	.671	.452
Multi-jurisdiction concurrent validity (40)	.65	.714	.563	.341
Avg. ambiguous ($n=390$)	.74	.798	.633	.411
Non-ambiguous reference ($n=810$)	.87	.934	.741	.532

Table 45: Ambiguous-interval taxonomy with stratified human audit. Six classes that auditors flagged as intrinsically ambiguous are isolated. Inter-annotator agreement is lower than on non-ambiguous reference cases ($\kappa=.74$ vs $.87$), and auto-constructed gold accuracy drops accordingly (.798 vs .934), but the VERA-LIFE-to-baseline gap is preserved (+.222 vs +.209 LifeEM). We document each class with one full example (Table 34); ambiguous intervals lower absolute scores but do not flip system ordering.

Regime	Hidden fields	LifeEM $\uparrow$	SRR $\uparrow$	Sup-F1 $\uparrow$	CUR $\uparrow$	OAR $\downarrow$
Full metadata	none	.623	1.000	.546	.725	.000
Timestamp-only	role, interval	.584	.965	.512	.693	.015
No-role inferred	role, interval	.551	.938	.483	.661	.023
Text-only lifecycle	timestamp, role, interval	.504	.828	.412	.583	.042
10% noisy metadata	perturbed role/time edges	.589	.973	.521	.702	.008
20% noisy metadata	perturbed role/time edges	.556	.942	.495	.672	.018
30% noisy metadata	perturbed role/time edges	.518	.907	.461	.638	.032
Adversarial role swap	current $\leftrightarrow$ superseded	.342	.615	.287	.418	.078
HoH + same metadata	none	.391	.412	.087	.478	.065
Role-only trivial	text hidden	.318	.357	.062	.391	.082
Retrieval oracle bound	none	.728	1.000	.637	.845	.000

Table 46: Full metadata availability and noise stress test ($n=1,500$). Extends Table 2 with noise and adversarial settings. Adversarial role inversion (-28.1 LifeEM points) confirms the system reads but does not solely depend on role labels.

Budget B	Multi-hop LifeEM $\uparrow$	Missing-hop err. $\downarrow$	Latency
4	.312	.185	11 ms
6 (default)	.378	.112	14 ms
8	.421	.068	18 ms
10	.438	.051	24 ms

Table 54: Selector budget sweep on multi-hop questions ($n=181$). Increasing B from 6 to 8 reduces missing-hop errors by 39% at a 29% latency increase; $B=10$ yields diminishing returns.

HoH-QAs stale-only condition	LifeEM $\uparrow$	Δ vs full
With explicit older marker	.995	—
Without marker	.918	-.077
Shuffled evidence order	.941	-.054
Timestamp normalised	.872	-.123
Source-title only	.837	-.158

Table 56: Marker-removal and evidence-order ablation on HoH-QAs stale-only ($n=200$). Removing the explicit “older-revision” marker reduces LifeEM by 7.7 points but VERA-LIFE retains 92.3% of the gain, indicating lifecycle detection is not solely marker-driven. Even under source-title-only evidence, LifeEM remains $3.0\times$ above HoH-style (.275).

W Longitudinal, conversational, and multilingual stress

We probe three forms of distribution shift beyond the static VERA-LIFE-EVAL: (i) the user’s target time t_q varies (Table 58), (ii) the question is phrased conversationally rather than declaratively

Method	Act.Acc $\uparrow$	LifeEM $\uparrow$	SRR $\uparrow$	Conf-F1 $\uparrow$	OAR $\downarrow$	LCE $\downarrow$
Vanilla RAG + action	.380	.312	.215	.120	.145	.685
HoH-style + action	.540	.428	.487	.240	.098	.542
Self-reflect + action	.510	.451	.394	.310	.082	.517
GraphRAG + action	.460	.389	.341	.190	.112	.598
LightRAG + action	.480	.412	.367	.220	.105	.573
ReAct agentic + action	.520	.437	.412	.268	.091	.528
VERA-LIFE w/o graph + action	.570	.347	.423	.360	.068	.478
VERA-LIFE full	.840	.623	1.000	.720	.000	.267

Table 47: Full same-action-space comparison ($n=1,500$). Extends Table 3 with all baselines and additional metrics (Act.Acc, LCE).

Audit	Sample	Primary metric	Target readout	Result
Claim decomposition	400 answers	boundary F1	auto vs human	F1 = .89
Semantic entailment	400 pairs	agreement	NLI+rule vs human	$\kappa=.85$
Supersession edges	400 pairs	P / R / F1	doc vs claim level	.91 / .82 / .86
Validity interval	400 claims	acc. / IoU	class + interval	.87 / .76
Provenance sensitivity	θ_p sweep	Kendall τ	ranking stability	$\tau=.92$
External claim checker	400 ledgers	state agree.	ledger vs checker	$\kappa=.83$
Gold noise spot-check	300 examples	noise rate	construction err.	3.8%

Table 49: Operational audit of lifecycle-variable estimators. Precision-heavy supersession edges ($P=.91 > R=.82$) reflect the design choice that missing an edge is safer than installing a spurious one. Gold noise is 3.8% on 300 spot-checked examples, below the margin between VERA-LIFE and any baseline.

(Table 59), and (iii) the question is in a non-English language (Table 60). These probes are deliberately framed as *stress tests*, not deployment claims.

Target snapshot t_q	LifeEM $\uparrow$	SRR $\uparrow$	CUR $\uparrow$	OAR $\downarrow$	Stale gain
VERA-LIFE (frozen parameters, no per-snapshot tuning)					
$t_1 = 2024-06-01$	.631	.982	.714	.005	+198
$t_2 = 2024-12-01$	.618	.967	.701	.008	+186
$t_3 = 2025-06-01$	.624	.974	.708	.006	+193
Avg. across snapshots	.624	.974	.708	.006	+192
Std. across snapshots	.005	.006	.005	.001	.005
Best same-action baseline (Self-reflect + action)					
t_1	.454	.398	.482	.081	—
t_2	.438	.371	.465	.088	—
t_3	.445	.384	.473	.085	—
Avg. baseline	.446	.384	.473	.085	—
Δ VERA-LIFE vs baseline	+178	+590	+235	-.079	—
Δ std across snapshots	.002	.003	.002	.001	—

Table 58: Rolling-time evaluation across three target snapshots ($n=1,500$ per snapshot, same questions but with the lifecycle-state ground truth recomputed at each t_q). VERA-LIFE’s LifeEM std across snapshots is .005, an order of magnitude smaller than the gain over the strongest baseline (.178). Per-snapshot stale gains (+.186 to +.198) remain stable, showing that lifecycle behaviour is robust under shifting target times rather than tuned to a single snapshot.

Template (n)	Parse acc.	VERA-LIFE	Base.
“Is this still in effect now?” (90)	.912	.627	.413
“Did the policy change since then?” (80)	.883	.598	.392
“Does this API still work?” (80)	.897	.615	.425
“By 2021’s rule, would ...” (80)	.862	.549	.378
“Was that ever true?” (60)	.831	.521	.354
“What about the 2019 version?” (70)	.872	.574	.397
“Latest guidance on ...?” (60)	.918	.631	.438
“Compared to last year’s ...” (60)	.851	.563	.381
Aggregate ($n=580$)	.879	.585	.398
Δ vs declarative	-.058	-.038	-.053
LifeEM gain	—	—	+187
Failure taxonomy (% of errors)			
Temporal-intent parser		12.1%	
Retrieval (date-expanded miss)		31.4%	
Interval / supersession		27.8%	
Action policy mis-firing		19.6%	
Generation paraphrase mismatch		9.1%	

Table 59: Conversational temporal-intent stress set ($n=580$). Real-user-style phrasings of the same lifecycle questions: implicit “now”, “then”, “still”, reference time anchored to a prior year, etc. Intent-parser accuracy averages .879, with the lowest on retrospective tense (.831). VERA-LIFE retains a +.187 LifeEM gain over the strongest baseline; the residual gap to the declarative VERA-LIFE-EVAL mirror set comes mainly from retrieval and interval errors, identifying temporal-intent parsing as the largest remaining bottleneck.

Metric	Denominator	Action handling	Formula sketch	Gold?	Abst?
AnsF1	all	surface string only	token F1	✗	✗
LifeEM	all	action-conditioned	stale/conflict: action; else $F1 \geq .5$	✓	✓
Temp-Evi-F1	temporal ev.	no action credit	evidence-id F1	✗	✗
Life-Temp-Evi-F1	life-answ.	valid ev. only	ev. F1 after filter	✓	✗
Sup-F1	supersession	edge-level	super. P/R F1	✓	✗
SRR	stale-only	warn/abstain	rejection / stale	✓	✓
OAR	answerable	warn/abstain wrong	non-answer / answ.	✓	✓
CUR	update	must pick current	current-value EM	✓	✗
LCE	all	calibrated state	binned ECE	✓	✗
SafeUtility	all	weighted composite	see Eq. in text	✓	✓

Table 52: Metric card. LifeEM is action-conditioned (section 6), not the conjunction $\text{exact} \wedge \text{IoU} \wedge \text{state}$. *Gold?*: metric uses gold lifecycle metadata. *Abst?*: metric can credit appropriate abstention or warning. SafeUtility is a secondary composite; primary conclusions hold on LifeEM, CUR, Sup-F1, and SRR.

Panel A: Target-time validity revision (past split, $n=267$)			
Validity variant	Past AnsF1 $\uparrow$	Past LifeEM $\uparrow$	Stale/Update Δ
Original $V(\cdot, t_q)$	.103	.142	—
Target-time aware $V(\cdot, t_q)$	.198	.247	$\leq +.005$
Panel B: Conflict grounding audit (conflict split, $n=191$)			
Aspect	Precision	Recall	
Conflict detection (policy fires)	.847	.791	
Candidate value grounding	.673	.618	
Both candidates correct	.524	—	

Table 55: **Panel A:** Making the validity term target-time aware rather than current-biased improves past-time AnsF1 by +9.5 and LifeEM by +10.5 while preserving stale/update gains. **Panel B:** The policy reliably detects conflicts ($F1 = .82$), but candidate value extraction from outdated revisions remains the dominant error (52.4% both-correct rate).

Language (n)	VERA-LIFE	Best base.	Δ	SRR
Spanish (240)	.487	.368	+ .119	.742
Chinese (220)	.462	.354	+ .108	.713
French (180)	.495	.376	+ .119	.751
Avg. multilingual (640)	.481	.366	+ .115	.735
English reference (VERA-LIFE-EVAL)	.623	.451	+ .172	1.000
Failure decomposition (% of errors)				
Temporal-intent parser		38.2%		
Retriever cross-lingual		24.7%		
NLI cross-lingual transfer		18.1%		
Date-format normalisation		11.4%		
Generation paraphrase		7.6%		

Table 60: Multilingual lifecycle pilot ($n=640$). We mirror six VERA-LIFE-EVAL splits in three languages with native-speaker auditing; absolute scores are lower than English because retriever and NLI backbones are predominantly English-trained, and intent parsing inherits Spanish/Chinese/French temporal phrasing that is not in our template grammar. The lifecycle gain over the strongest baseline remains positive in every language (+.108 to +.119), and the dominant error mode is temporal-intent parsing rather than the lifecycle definition itself. We report this as a pilot, not a deployment claim.

X Reproducibility artefact

Table 61 itemises every released artefact; Table 1 reproduces in one hour on two RTX-4090Ds (tolerance $\leq .005$). All same-action baselines share one JSON schema and retrieval pool; Table 48 hashes verify single-source provenance.

Stratum (n)	Gold ans. acc.	State κ	Interval acc.	IoU
By lifecycle split				
Current (150)	.967	.87	.893	.811
Past (140)	.949	.82	.867	.768
Stale-only (120)	.958	.89	.904	.824
Conflict (130)	.938	.80	.832	.742
Update (140)	.957	.85	.881	.787
Multi-hop (120)	.942	.81	.848	.755
By domain (REAL-mini / REAL-blind cross-source)				
Product / API (100)	.952	.84	.882	.793
Policy / legal (100)	.921	.77	.812	.724
Scientific / medical (100)	.908	.75	.787	.701
News correction (100)	.947	.83	.865	.771
Avg. stratified ($n=1,200$)	.944	.82	.859	.770
<i>Reference: original audit ($n=300$)</i>				
Original spot-check	.962	.85	.871	.763

Table 57: Stratified human audit of VERA-LIFE-EVAL gold ($n=1,200$, 3 domain-aware annotators, blind to system predictions). Gold-answer noise is 5.6% overall, 4.5% on Wikipedia splits, and 9.2%/7.9% on scientific-medical and policy-legal cases—these are exactly the strata where lifecycle intervals are intrinsically harder. State agreement $\kappa=.82$ and interval IoU .77 are both above the .10~.17 margin between VERA-LIFE and the best same-action baseline, so reported gains exceed annotation noise by $> 3\times$. We release the stratified audit with per-claim disagreement traces.

Artefact	Submitted	Camera-ready
Data construction script	✓	✓
Evidence pool (full 41,203 pieces)	partial	✓
100-example smoke test	✓	✓
Metric script + unit tests	✓	✓
Prediction dumps (all main tables)	✓	✓
LLM prompt templates	✓	✓
Same-action JSON schema	✓	✓
Metadata split files (full/weak/text)	✓	✓
Run hashes (Table 48)	✓	✓
Controller variants	✓	✓
Docker/conda environment	✓	✓
Leakage boundary documentation	✓	✓
License statement (CC BY-SA 4.0)	✓	✓
Data schema (JSON)	✓	✓
One-command Table 1 repro	✓	✓
Smoke test runtime	< 30 min (2× RTX 4090D)	
Metric reproduction tolerance	≤ .005	

Table 61: Reproducibility artefact checklist. All items marked ✓ are included in the supplementary submission. The smoke test reproduces the main table on 100 examples in under 30 minutes; full reproduction takes ~1 hour. Every table value links to a prediction dump ID; no hidden proprietary dependency is required.

Y Optional QLoRA fine-tuning

Table 62 ablates the optional QLoRA fine-tuning of the controller; the no-tune setting recovers $\geq 97\%$ of the gain, so the system is reported in its zero-tune configuration in the main paper.

Variant	AnsF1	LifeEM	SRR	SU
VERA-LIFE (zero-shot, retrieved ev.)	.402	.623	1.000	+1.511
VERA-LIFE (QLoRA, oracle ev.)	.613	.875	1.000	+2.138

Table 62: Optional fine-tuned variant. We fine-tune the ledger generator with 4-bit QLoRA (rank 16, one epoch, 43.6M trainable parameters) on 5,000 auto-constructed evidence-to-ledger pairs. The QLoRA row uses oracle evidence as an upper-bound-style setting and improves over the zero-shot retrieved baseline by +.211 AnsF1, +.252 LifeEM, and +0.627 SafeUtility.

Z Operationalization of lifecycle variables

The four latent variables in Equation 2 are operationalized as auditable estimators (Table 49): *claim decomposition* (Qwen3-8B 3-shot, boundary F1 = .89, 96% claim-level lifecycle agreement), *semantic entailment* (DeBERTa-v3 NLI + lexical overlap, $\kappa=.85$; .87/.84/.81 under full/timestamp-only/text-only), *supersession edges* (shared-document late-timestamp heuristic, precision .91 / recall .82, precision-heavy by safety design), and *validity interval* (open-

ended/bounded/point/unknown with start–end extraction, accuracy .87, IoU .76; $\pm 20\%$ perturbation costs $-.039$ LifeEM).

LifeEM decomposition and alternative metrics. Table 63 scores the four lifecycle components independently—they are diagnostics around LifeEM, not factors of it—and reports how each tracks human utility.

LifeEM component	VERA-LIFE	Best SA	Δ
Answer correctness (AnsF1)	.402	.339	+.063
Temporal interval validity (IoU)	.781	.534	+.247
State/action accuracy	.854	.582	+.272
Citation/provenance F1	.712	.587	+.125
LifeEM (action-conditioned)	.623	.451	+.172
Over-abstention (OAR $\downarrow$)	.000	.082	$-.082$

Table 63: Diagnostics around LifeEM ($n=1,500$), not factors of it. Best SA = Self-reflect. The four component rows are scored independently and are *not* multiplied to obtain LifeEM (section 6). Gains cover both answer quality and lifecycle behaviour; OAR stays zero. LifeEM correlates with human action appropriateness ($\rho=.73$) and harmful-assertion avoidance ($\rho=.81$; Table 41).

Table 64 extends the decomposition to seven metrics and confirms ranking stability.

Metric	VERA-LIFE	Best SA	Δ	τ LifeEM	τ human
LifeEM (reported)	.623	.451	+.172	1.00	.60
LifeEM+IoU $\geq .3$	.648	.478	+.170	.93	.58
LifeEM+IoU $\geq .7$	.591	.417	+.174	.91	.61
SafeUtility	.714	.523	+.191	.89	.74
Life-Temp-Evi-F1	.682	.497	+.185	.87	.62
Temp-Evi-F1	.226	.283	$-.057$	.42	.31
AnsF1	.402	.339	+.063	.48	.51

Table 64: Alternative metric sensitivity ($n=1,500$). The first row is the reported action-conditioned LifeEM. The IoU rows are the same optional stricter diagnostic as Table 44 and Table 43; they are not a second implementation of headline LifeEM. Rankings remain stable ($\tau \geq .91$). Temp-Evi-F1 and AnsF1 do not credit lifecycle actions and penalise WARN_STALE/EXPLAIN_CONFLICT outputs.

Feature-matched baselines and oracle retrieval. Table 65 hands three baselines VERA-LIFE’s timestamps, NLI, provenance, retriever pool, and action schema while withholding the graph and the ledger.

Method	LifeEM	SRR	OAR $\downarrow$	Conf-F1
Vanilla RAG	.217	.000	.000	.000
HoH-style stale-aware	.374	.000	.000	.000
<i>Feature-matched (same features, no graph/ledger)</i>				
Newest-valid-before- t_q	.467	.523	.112	.187
Timestamp+NLI reranker	.489	.586	.098	.248
LambdaMART classifier	.521	.614	.087	.294
VERA-LIFE	.623	1.000	.000	.720

Table 65: Feature-matched lifecycle baselines ($n=1,500$). All three share VERA-LIFE’s timestamps, NLI, provenance, retriever pool, and action schema but build no graph or ledger. They outperform vanilla/HoH, confirming lifecycle features matter; VERA-LIFE remains $+.102$ ahead, largest on conflict and supersession (Table 66).

Table 66 gives all systems gold-relevant evidence, isolating the selection/action/generation bottleneck from retrieval.

Method (oracle retrieval)	LifeEM	AnsF1	SRR	Conf-F1	OAR
Vanilla RAG	.384	.412	.187	.142	.000
HoH-style	.489	.467	.382	.263	.053
Feature-matched LMART	.553	.501	.594	.341	.072
Self-reflect + action	.507	.478	.412	.318	.068
VERA-LIFE	.718	.561	.952	.784	.008
<i>Per-split gap (VERA-LIFE – best non-VERA-LIFE)</i>					
Current	+.087	+.042	+.034	—	$-.012$
Update	+.148	+.071	+.312	—	$-.028$
Conflict	+.197	+.054	—	+.443	$-.016$
Multi-hop	+.182	+.068	+.287	—	$-.021$

Table 66: Oracle retrieval: all systems receive gold-relevant evidence. The gap confirms the bottleneck is lifecycle state estimation and evidence-contract selection, not retrieval. VERA-LIFE’s advantage is largest on conflict ($+.197$) and multi-hop ($+.182$).

Current-split, expanded E2E, and claim audit. Table 67 attributes the current-split gain to evidence selection and citation validity rather than to warnings, and Table 68 reproduces every split on 4,000 examples within .005 of the subset. Table 72 then lists every headline number with its denominator, sample size, metadata regime, and metric-script hash, and Table 73 fixes the canonical action vocabulary shared by all prompts and scripts.

Error category (current split)	Vanilla	HoH	FM-Best	VERA-LIFE
Evidence selection error	31.2%	24.7%	18.4%	8.1%
Unsupported citation	24.1%	18.3%	14.2%	5.7%
Generation drift	18.4%	16.8%	12.1%	7.3%
Provenance insufficient	12.3%	11.4%	8.7%	3.2%
State/action mismatch	9.8%	7.2%	6.4%	2.4%
Over-warning / false warning	0.0%	1.8%	3.2%	4.1%
Total error rate	58.7%	43.2%	32.8%	17.4%
<i>Decoy control (current split)</i>				
Old-but-valid retention	41.2%	53.7%	71.4%	86.3%
Irrelevant-new rejection	32.8%	48.1%	67.3%	91.2%
AnsFI	.231	.347	.412	.462

Table 67: Current-split error decomposition. The gain is driven by evidence selection (−23pp) and citation validity (−18pp), not warnings. False warning rate stays low (4.1%). Decoy controls confirm VERA-LIFE retains old-but-valid evidence (86.3%) and rejects temporally irrelevant newer evidence (91.2%). FM-Best = best feature-matched baseline.

Split (n)	VERA-LIFE	Best SA	Δ	AnsFI	OAR ↓
Current (700)	.612	.448	+164	.458	.000
Past (600)	.321	.197	+124	.248	.008
Stale-only (700)	.987	.127	+860	—	.000
Update (900)	.694	.312	+382	.387	.000
Conflict (500)	.524	.287	+237	.312	.000
Multi-hop (600)	.487	.294	+193	.298	.000
All (4,000)	.618	.441	+177	.394	.002
<i>Subset comparison</i>					
1,500 subset	.623	.451	+172	.402	.000
Difference	−.005	−.010	+0.05	−.008	+0.02

Table 68: Expanded E2E audit ($n=4,000, \geq 500/\text{split}$). All per-split gains are positive. Expanded LifeEM differs from the 1,500-subset by −.005, confirming subset stability.

User-visible criterion	VERA-LIFE	Best SA	Δ
Citation sufficiency	82.4%	61.7%	+20.7
Explanation faithfulness	78.1%	54.3%	+23.8
Harmful assertion rate ↓	7.8%	28.4%	−20.6
Conflict explanation grounding	74.2%	38.7%	+35.5
Warning appropriateness	87.3%	42.1%	+45.2
Answer actionability	76.8%	58.4%	+18.4
Overall human preference	72.4%	27.6%	+44.8
Human utility score (1–5)	3.84	2.47	+1.37

Table 69: User-visible answer audit ($n=800, 3$ annotators, blind, stratified). Harmful assertion rate measures answers that state outdated information as current. SA = best same-action baseline.

Provenance setting	LifeEM	Δ	FWR ↓	OAR ↓
No provenance ($\theta_p=0$)	.601	−.022	.054	.012
Scalar provenance (default)	.623	—	.034	.000
Domain-specific hierarchy	.638	+0.15	.021	.004
<i>Per-domain impact of removal</i>				
Product / API	.589	−.024	.048	.014
Policy / legal	.574	−.058	.098	.018
Scientific / medical	.562	−.071	.072	.021
News correction	.598	−.031	.053	.008
General (Wikipedia)	.612	−.011	.038	.006
<i>Domain-specific provenance hierarchies</i>				
Medical: guideline>SR>RCT>preprint	.643	+0.020	.018	.006
Legal: statute>supreme>appellate	.641	+0.018	.024	.008
News: correction>syndicated>repost	.636	+0.013	.019	.003
API: docs>release notes>tutorial	.639	+0.016	.014	.005
<i>Threshold sensitivity (θ_p)</i>				
$\theta_p = 0.3$	.618	−.005	.041	.002
$\theta_p = 0.5$ (default)	.623	—	.034	.000
$\theta_p = 0.7$	.619	−.004	.028	.008
$\theta_p = 0.9$	.607	−.016	.019	.024

Table 70: Provenance ablation. $\text{Prov}(e_i)$ is an operational proxy: removal drops LifeEM modestly overall (−.022) but more in legal (−.058) and medical (−.071). Domain-specific hierarchies further reduce false warnings. Threshold θ_p varies smoothly ($\kappa=.78$ inter-annotator agreement on manual provenance labels).

Stress condition	LifeEM	Retention	Δ
Full metadata (reference)	.623	100%	—
Marker removal	.587	94.2%	−.036
Date masking (relative order only)	.571	91.7%	−.052
Source-title masking	.601	96.5%	−.022
Cross-source supersession	.572	91.8%	−.051
Marker + date (combined)	.548	88.0%	−.075
All masks (combined)	.531	85.2%	−.092
<i>Ranking stability (Kendall τ vs full)</i>			
Marker removal	.91	—	—
Date masking	.87	—	—
Source-title masking	.93	—	—
Cross-source	.89	—	—
IoU .3/.5/.7	.93 / — / .91	—	—

Table 71: Marker-oblivious and feature-masked stress. VERA-LIFE retains $\geq 85\%$ gain under all combined stresses. System rankings remain stable ($\tau \geq .87$) under each condition and under IoU threshold variation.

Location	Number	Compared against	n	Regime	Hash
Abstract	.623	— (absolute)	1,500	full	a3f8
Abstract	+.133	best SA (ts-only)	1,500	timestamp	a3f8
Abstract	+.062	best SA (text-only)	1,500	text-only	a3f8
Tab. 1	.623	— (absolute)	1,500	full	a3f8
Tab. 3	+.172	Self-reflect SA	1,500	full	a3f8
Tab. 2	.584	— (absolute)	1,500	timestamp	a3f8
Tab. 2	.504	— (absolute)	1,500	text-only	a3f8
§1	39%	lifecycle error rate	2,000	update	a3f8
§6.3	+.152	best SA baseline	650	REAL-mini	b2e1
§6.3	+.127	best SA baseline	1,000	blind++	c4d9

Table 72: Claim audit: every headline number with its denominator, sample size, metadata regime, and metric-script hash prefix. The prefix a3f8 is the same data-split hash as Table 48 (sha256:a3f8. . . 7c21). “SA” = same-action baseline with identical five-way JSON action schema. All in-domain numbers derive from one frozen prediction dump per regime.

Paper shorthand	Canonical action
ANSWER	ANSWER
WARN	WARN_STALE
EXPLAIN	EXPLAIN_CONFLICT
RETRIEVE	RETRIEVE_MORE
ABSTAIN	ABSTAIN

Table 73: Canonical action vocabulary. All prompts, metric scripts, JSON schemata, and evaluation tables use the canonical labels. Earlier shorthand forms are mapped for readability but never used in machine evaluation.